

Store Less, Search Better: Access-Controlled Vector Search over Resolved Visibility

Chenyang Wu¹ Renxuan Wang² Zhongle Xie¹ Meihui Zhang³ Gang Chen¹

¹Zhejiang University ²Northwestern Polytechnical University ³Beijing Institute of Technology
 {cy.wu, xiezl, cg}@zju.edu.cn wangrenxuan@mail.nwpu.edu.cn meihui_zhang@bit.edu.cn

ABSTRACT

Vector databases increasingly underpin retrieval-augmented generation, recommendation, and multi-agent applications. In these systems, queries from different users may operate over different visible subsets of a shared collection. A shared index limits replication but may waste substantial search effort on invisible data, whereas isolated collection partitions provide cleaner search spaces at the cost of replicated storage. Yet existing partitioning methods are often designed around a particular authorization policy, which are expensive to rebuild as visibility evolves.

We present SQUID, a policy-independent framework that performs physical design over resolved user-data visibility. SQUID groups items with identical visible-user sets into Access Atoms and places them into memory-bounded partitions. A Shared-Atom Graph and a recall-aware cost model balance storage savings against query latency. SQUID further allocates search effort according to route difficulty and enforces current visibility before returning results. When data or visibility changes, it updates logical visibility immediately and repairs only the affected physical state. Experiments across multiple datasets and authorization policies show that SQUID achieves lower latency while using less memory than state-of-the-art methods.

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PVLDB Artifact Availability:

The source code, data, and/or other artifacts have been made available at <https://github.com/zjuDBxAI/SQUID>.

1 INTRODUCTION

Vector databases increasingly support retrieval-augmented generation, recommendation, and multi-agent applications [18, 20, 37]. These deployments rely on a broad range of approximate nearest-neighbor (ANN) algorithms and systems, whose performance varies across datasets, workloads, and constraints [3, 7, 31]. In these applications, application-level access constraints give users different visibility over vectors embedded in a common semantic space. We

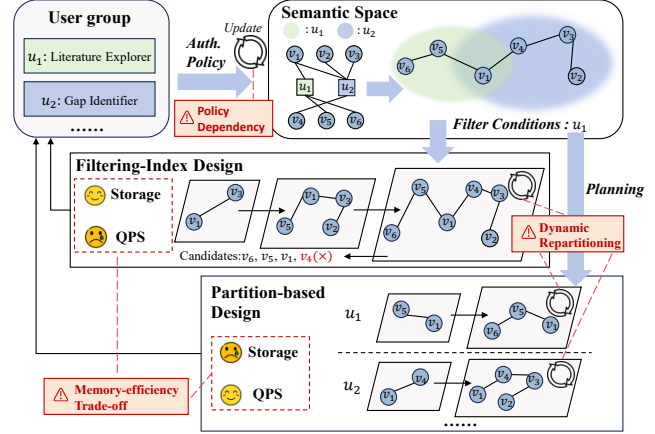


Figure 1: An access-controlled multi-agent system. The system faces three design challenges: policy dependency in deriving visibility, the memory-efficiency trade-off of storage organization, and dynamic repartitioning after policy updates.

call the resulting task *access-controlled vector search*: a query must retrieve approximate nearest neighbors only from the vectors visible to its user. A vector outside this valid search domain is unauthorized, or *invisible*, rather than an approximate answer, even if it is close to the query vector. The same vector may be visible to several users, and each user’s visible vectors change as vectors and visibility are updated over time. The vector database must therefore account for different visible subsets in both storage organization and search execution to ensure efficiency.

Figure 1 illustrates this problem in an AutoResearch agent system. Two users with roles, i.e., Literature Explorer and Gap Identifier, access overlapping but nonidentical sets of vectors embedded in the same semantic space. A neighborhood that is valid for one user to search may include vectors invisible to the other. Although the queries for different users use the same similarity measure, they induce different search spaces. External authorization policies, including role-based access control (RBAC) [8, 19, 30], attribute-based access control (ABAC) [5, 14], and discretionary access control (DAC) [1], determine which vectors each user may access. We call the resulting user–vector mapping *resolved visibility*.

This setting raises a database design question: *How should a vector database organize and search its stored vectors when each user has different visibility over them?* The central difficulty is that keeping storage compact and giving each user a focused search space favor different organizations. Keeping shared vectors together avoids storing them repeatedly, but a query may encounter many vectors

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that its user cannot access. Separating vectors by user visibility narrows each user’s search space, but vectors must appear in several places. Because visibility differs across users and changes over time, improving storage or search in isolation can make the other less efficient.

Existing systems have consequently moved visibility enforcement from query-time filtering toward the physical organization of searchable data, as illustrated in Figure 1. Filtered-Index design maintains a shared index over the stored vectors and applies visibility constraints during search [12, 22, 27, 45]. Albeit compact in storage, filtered ANNS can incur substantial query computation on invisible vectors. The resulting computation on invisible vectors can increase end-to-end query latency and reduce throughput, particularly when a user’s valid search domain is sparse.

To reduce this query-time filtering, partition-based designs further materialize visibility-aligned search spaces. Partition-based designs materialize smaller searchable partitions and route each query to partitions that may contain visible vectors [13, 22, 33, 43, 44]. By directing each user toward cleaner permitted search spaces, partitioning can avoid much of the invisible-vector work incurred by filtered ANNS. However, vectors visible to multiple users may be stored and indexed repeatedly. Partition planning therefore couples two decisions: which vector copies to retain in storage and which partitions each user must search. Moreover, when a user searches multiple partitions, the database must coordinate those searches and combine their candidates into one result. This evolution makes *partition planning*, which is the process of constructing partitions, vector copies, and user routes, i.e., the partitions searched for each user, a core design task for partition-based methods. Yet, existing state-of-the-art methods [13, 26, 44] face three challenges.

Challenge 1: policy-independent partition planning. Existing methods commonly perform partition planning over roles, attributes, or other authorization-policy objects, making the physical organization depend on the representation that produced resolved visibility. When these policy objects change, the derived partitions and indexes may need to be rebuilt. Prior work reports that such rebuilding can take hours on 1M vectors [44], making repeated construction an adaptation bottleneck. Planning separately for every vector removes this dependence but makes the decision space grow with the collection. Partition planning must therefore derive tractable decision units from resolved visibility without relying on either policy objects or individual vectors.

Challenge 2: balancing memory consumption and query computation. Partition-based methods face an inherent trade-off in how much user-specific isolation to materialize: greater isolation reduces computation on invisible vectors but replicates shared vectors, whereas sharing partitions among users saves memory but increases query computation. A vector copy or user route can change the search spaces of several users, so these decisions cannot be optimized per user or per partition. Moreover, planned partitions differ in size and in the fraction of vectors visible to each routed user, so the same query computation can yield different recall across partitions. Under a memory budget, partition planning must therefore compare each copy’s memory consumption with the query computation it saves across the affected user-partition pairs at a common recall target.

Challenge 3: maintaining evolving partition organizations.

Materializing the selected partitions, copies, and routes produces a *partition organization* whose indexes and metadata can become misaligned with users’ current valid search domains as vectors or resolved visibility change. This misalignment can lower recall or increase query computation until the affected partitions and indexes are repaired. During repair, query execution must respect current resolved visibility and maintain the target recall, while copies, routes, partition contents, indexes, and metadata are updated consistently. Maintenance must therefore coordinate these repairs and restore query efficiency without making every change trigger database-wide repair.

We present SQUID, a framework that performs database-internal physical design over resolved visibility for access-controlled vector search. SQUID groups vectors with the same visible-user set into policy-independent *Access Atoms*. Starting from a selective per-user seed, its sparse Shared-Atom Graph exposes redundant Atom copies, and its recall-aware cost model ranks their consolidation under a memory budget. At query time, SQUID allocates partition-specific search effort, filters candidates under current resolved visibility, and merges authorized candidates across routed partitions. When vectors or visibility change, SQUID preserves current query correctness while repairing the affected partitions and indexes.

Contributions. Our contributions are as follows:

- We formulate access-controlled vector search as a design problem over resolved visibility, and devise Access Atoms as policy-independent units connecting authorization semantics to partition planning, search and maintenance. (§2 & §3)
- We design a memory-bounded Partition Planner that exposes redundant copies through a Shared-Atom Graph and ranks their consolidation at a certain recall target. (§4)
- We develop a partition-aware Search Engine and a maintenance protocol that serves current visibility while repairing affected storage, indexes, and routing metadata. (§5)
- We evaluate SQUID’s advance across multiple datasets, authorization policies, memory budgets, and update workloads against SOTA filtered-index and partition-based alternatives. (§6)

2 SYSTEM MODEL AND RETHINKING

In access-controlled vector search, each query is evaluated over the subset of vectors visible to its user. Resolved visibility determines this valid search domain, while query efficiency depends on how the vector database organizes and searches the underlying vectors. This section first formalizes the system model, then revisits how resolved visibility shapes storage organization, index and search, and maintenance, and finally derives the design principles that guide SQUID.

2.1 System Model

Let $U = \{u_1, \dots, u_{|U|}\}$ and $D = \{d_1, \dots, d_{|D|}\} \subseteq \mathbb{R}^h$ denote the user set and vector collection, respectively, where h is the vector dimensionality. The visibility of vectors to users is resolved by one or more authorization policies [5, 8, 14, 30]. We model the resolved visibility by the function $\mathcal{V} : D \rightarrow 2^U$, where $u \in \mathcal{V}(d)$ if and only if vector d is visible to user u . The visibility function $\mathcal{V}$ is a

logical model; implementations that produce the same mapping from vectors to visible-user sets are equivalent under this model.¹

An access-controlled vector-search query is a tuple $Q = (q, u, k)$, where $q \in \mathbb{R}^h$ is a query vector, $u \in U$ is the querying user, and $k \in \mathbb{N}^+$ is the requested result cardinality. The valid search domain of Q is $D_u = \{d \in D \mid u \in \mathcal{V}(d)\}$. The search returns an approximate top- k result $R(Q) \subseteq D_u$ under the chosen distance metric. Approximation may affect which near neighbors in D_u are retrieved, but every returned vector must remain in D_u ; a vector outside D_u is unauthorized rather than approximate.

The definitions of $\mathcal{V}$ and D_u specify query-visible vectors but leave the database realization open. A database must decide how vectors are organized into storage partitions, how ANN indexes are built over the partitions, how a query searches the resulting indexes, and how partitions, indexes, and auxiliary metadata are maintained against change [9, 10, 15, 17, 25, 32, 33]. Different realizations of the same $\mathcal{V}$ redistribute storage overhead, search computation, and maintenance work without changing the valid result domain D_u . The next subsection examines this design space in depth across storage organization, index and search, and maintenance.

2.2 Rethinking Design Space

To rethink this design space, we distinguish the planning process, its materialized organization, the indexes and search spaces that organization creates, and the maintenance required as visibility evolves. Note that *Partition planning* generates partitions, vector copies, and user routes, producing a *partition plan*. Materializing this plan produces a *partition organization* containing the stored copies, ANN indexes, routes, and associated metadata.

Storage organization. For partition planning, the relevant information is not which authorization-policy object grants access, but whether vectors have the same visible-user set. Vectors that are indistinguishable under resolved visibility can be considered together without changing any user’s valid search domain, reducing the granularity of the planning space. Physical decisions over these groups nevertheless remain coupled: removing a copy saves its storage once, but changes every user route that relied on it, and the resulting query computation depends on the partition into which its vectors are consolidated. The benefit of a copy is therefore determined by the surrounding partition organization rather than by the copy alone.

Index and search. Partition planning changes not only the number of indexes searched by a user, but also the size and visible-vector fraction of each searched partition. These properties affect the recall obtained from a given amount of query computation. Consequently, evaluating alternative organizations at a fixed search depth can favor a consolidation merely because it attains lower recall. Their query costs must instead be normalized by the computation required to reach a recall target. Given the selected organization, query execution then distributes this computation across the routed indexes and filters their candidates under current visibility.

Maintenance. Visibility changes affect serving correctness and physical quality differently. A revoke can be enforced by filtering candidates under current visibility even while an existing index

Table 1: Technical-route comparison.

Method	Storage organization		Index and search		Maintenance		Performance	
	Policy agnostic	Memory bounded	Recall-aware	Partition-aware	Delta update	Perf. repair	Mem. eff.	Query eff.
HONEYBEE [44]	×	✓	✓	✓	✓	×	Low	Low
VEDA/EFFVEDA [13]	×	✓	×	✓	✓	✓	Low	High
SQUID (Ours)	✓	✓	✓	✓	✓	✓	High	High

remains searchable, whereas a grant may require adding a reachable copy and route before the newly visible vectors can be found. The database must therefore update the affected copies, routes, and indexes consistently enough to serve the new valid search domains before optimizing their physical organization. Subsequent local reorganization can then remove redundant copies and recover recall and query efficiency within the affected region, avoiding reconstruction work proportional to the full database.

2.3 Design Principles

The three layers are coupled through visibility materialization. Storage decisions determine the indexes and search spaces exposed to each user; their query cost determines whether the memory consumed by a copy is justified; and visibility changes determine how long that organization remains useful and how much work is required to adapt it. Optimizing any layer in isolation can therefore shift cost to the other two. This fact yields three design principles for access-controlled vector search:

- **Policy-agnostic organization.** The database should derive tractable decision units from resolved visibility, avoiding dependence on both authorization-policy objects and vector-level decisions.
- **Benefit-driven materialization.** The planner should retain a materialized copy only when the query computation it saves across affected valid search domains at a common recall target justifies its memory and index overhead.
- **Resilient online maintenance.** Subsequent queries should observe current visibility while local repair restores recall and query efficiency in the affected partitions and indexes.

Discussion. As summarized in Table 1, representative state-of-the-art systems address complementary subsets of these three principles. They primarily follow a *policy-derived planning* route, using roles or role combinations to determine the partitions and copies considered during planning. HONEYBEE [44] exemplifies this route by optimizing role-derived partitions under memory and recall constraints. VEDA and EFFVEDA [13] couple memory-bounded organization with coordinated partition search and delta maintenance over role combinations. The resulting partitions and copies nevertheless remain tied to the policy representation, and maintaining a correct organization incrementally does not necessarily restore query performance after substantial drift. We instead follow a *visibility-driven planning* route: resolved visibility determines which vectors are equivalent for planning, copy benefits are measured at the same recall target, and the same equivalence localizes subsequent repair. The following sections define our abstraction, develop memory-bounded planning over it, and present the corresponding search and maintenance procedures.

¹We use *visibility* and *access* interchangeably throughout this paper.

3 SQUID

SQUID turns resolved visibility into a physical organization for access-controlled vector search. Its design centers on Access Atoms—groups of vectors with the same visible-user set. These Atoms provide a common unit across three stages: the Planner selectively replicates and consolidates them under a memory budget, the Search Engine searches the resulting partitions for each user, and the Maintainer repairs the affected state as vectors or visibility change. We first introduce this visibility-preserving unit and then follow it through SQUID’s offline, query, and update paths.

3.1 Key Idea

Physical organization needs a unit that preserves visibility semantics without depending on authorization syntax: individual vectors are too fine-grained for sharing detection, memory accounting, and maintenance, whereas roles or attributes bind the organization to the policy representation that produced visibility. SQUID therefore groups vectors by *shared visibility*. Under $\mathcal{V}$, vectors with the same visible-user set are indistinguishable for access-control correctness and form an Access Atom:

Definition 3.1. (Access Atom) An Access Atom for visible-user set $A_i \subseteq U$ is $S_{A_i} = \{d \in D \mid \mathcal{V}(d) = A_i\}$. All vectors in S_{A_i} are visible to exactly the same users. In Figure 2, v_1 belongs to the shared Atom of the Literature Explorer and Gap Identifier, while their private notes belong to different Atoms.

Discussion. Access Atoms are logical planning units rather than physical indexes. SQUID stores their copies in searchable *partitions*, where a partition may contain multiple Atoms and an Atom may appear in multiple partitions. Retaining copies yields smaller, cleaner user-visible search spaces; consolidating them saves memory but can create mixed partitions that require more filtering during search. Thus, unlike a global filtered index or policy-specific partitioning, Access Atoms expose the replication–search trade-off directly from resolved visibility and carry it consistently into planning, serving, and maintenance.

3.2 Architecture and Overview

Figure 2 organizes SQUID around the Access Atom Catalog produced from resolved visibility. The **Partition Planner** converts the catalog into memory-bounded Atom copies, searchable partitions, and per-user routes stored in the User Routing Directory. The Atom Inverted Table records the partitions containing each Atom, while the Shared-Atom Graph identifies partitions that share removable copies. The **Search Engine** consumes the routes, and the **Partition Maintainer** updates the same catalog, placement, routing, graph, and index state after vector or visibility changes.

Process and Data Flow. The three paths carry the same Atom-level state. Offline, the Atomizer converts resolved visibility into catalog entries, and the Planner produces memory-bounded partitions and records each user’s route. In Figure 2, the shared Atom can initially be copied in both user-local partitions; the Planner subsequently decides whether to retain or consolidate these copies under the memory budget. At query time, the Search Engine follows the requesting user’s route, searches its partitions, filters candidates against current visibility, and merges the authorized top- k . On an

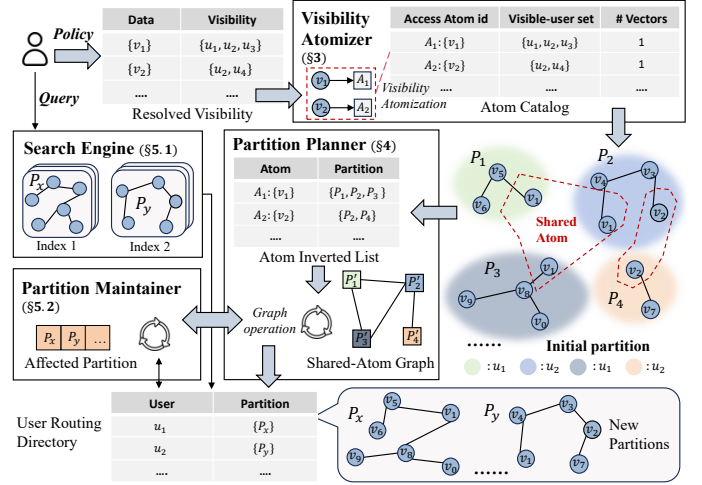


Figure 2: SQUID Overview

update, the Maintainer changes the affected visibility-equivalent group and repairs only the corresponding catalog, placement, route, graph, and index state rather than rebuilding the full organization. **Example.** For the Literature Explorer u_1 and Gap Identifier u_2 , the Atomizer places their shared vector v_1 in A_1 and keeps their remaining visibility groups as separate Atoms. The Planner consolidates the initial $P_1, \dots, P_4$ into P_x and P_y ; the User Routing Directory records the partitions to search for each agent. The Search Engine uses these entries to search authorized data, while the Maintainer updates the affected Atom, partition, and directory state when data or visibility changes.

4 PARTITION PLANNING

Before planning, the Visibility Atomizer groups vectors with identical resolved visible-user sets into Access Atoms and records them in the Access Atom Catalog. Given this catalog, the Partition Planner constructs a partition plan and user routes under a memory budget and target recall by exposing redundant copies, generating coverage-preserving consolidations, and ranking them with the recall-aware cost model in §4.4.

4.1 Partition-Planning Problem

The catalog specifies which vectors can move together, but not how their copies should be organized for search. The Partition Planner takes the catalog, a target $\text{Recall}@k$, and memory budget $\mathcal{B}$ as input. It must keep every visible vector reachable, keep the vectors stored across materialized partitions within $\mathcal{B}$, and minimize query computation at the target recall.

A *partition* P is a searchable unit containing Atoms $\mathcal{S}(P)$, with vector cardinality $|P|$; an Atom may be copied across partitions. A *partition plan* $\mathcal{P}$ comprises the partitions, vector-copy placements, and user routes generated by partition planning, and a user’s route $\mathcal{P}(u) \subseteq \mathcal{P}$ is the set of partitions it searches. Every route must provide *route coverage*:

$$D_u \subseteq \bigcup_{P \in \mathcal{P}(u)} P, \quad \forall u \in U. \quad (1)$$

Coverage ensures reachability, not route purity: a routed partition may contain Atoms invisible to its user, whose candidates are filtered during search. Planning therefore trades the memory saved by removing redundant copies against the extra query computation of searching mixed partitions.

Memory consumption. Partition planning is subject to an end-to-end memory limit covering vector storage, ANN indexes, and metadata. Because exact index and metadata overhead is unavailable before materialization, SQUID controls its dominant planning-dependent component—vector copies, i.e., physical occurrences of vectors in materialized partitions:

$$M(\mathcal{P}) = \sum_{P \in \mathcal{P}} \sum_{S_{A_i} \in \mathcal{S}(P)} |S_{A_i}|. \quad (2)$$

The Planner therefore enforces the vector-copy budget $\mathcal{B} = \beta|D|$, where $\beta \geq 1$, while §6.3 reports the realized end-to-end memory after materialization.

Query computation. Query computation is incurred whenever a user’s route searches a partition. Let $C(u, P)$ denote the query computation required for u to search routed partition P at the target Recall@ k . The plan-level query computation sums this cost over all user–partition pairs:

$$C(\mathcal{P}) = \sum_{u \in U} \sum_{P \in \mathcal{P}(u)} C(u, P). \quad (3)$$

§4.4 gives the concrete definition of $C(u, P)$; the abstract form above is sufficient for the planning problem.

Definition 4.1 (Memory-Bounded Partition Planning (MBP²)). Given the Access Atom Catalog, memory budget $\mathcal{B}$, target Recall@ k τ , and per-route query computation $C(u, P; \tau)$, MBP² is

$$\begin{aligned} \min_{\mathcal{P}, \{\mathcal{P}(u)\}_{u \in U}} \quad & C(\mathcal{P}) \\ \text{s.t.} \quad & M(\mathcal{P}) \leq \mathcal{B}, \\ & D_u \subseteq \bigcup_{P \in \mathcal{P}(u)} P, \quad \forall u \in U. \end{aligned} \quad (4)$$

Here, the target recall τ is an explicit input to MBP², and $C(u, P; \tau)$ denotes the per-route query computation at that target; we write $C(u, P)$ when τ is fixed.

The decision version of MBP² is NP-hard; Appendix C gives its formal statement and proof. Because globally coupled copy-removal choices do not scale to many Atoms and users, SQUID starts from a coverage-preserving seed and iteratively applies locally generated consolidations until the budget is met.

Running example. Figure 3 starts with four partitions containing two replicated Atoms. The Atom $A_1 = \{v_1\}$ is owned by P_1, P_2, P_3 , while $A_2 = \{v_2\}$ is owned by P_2, P_4 . These placements contribute three copies of A_1 and two copies of A_2 to $M(\mathcal{P})$. The following stages use their owner sets to expose candidate edges and select consolidations that preserve route coverage.

4.2 Shared-Atom-Based Candidate Generation

SQUID initializes planning with one partition $P_u^{(0)} = D_u$ and one route $\mathcal{P}^{(0)}(u) = \{P_u^{(0)}\}$ for each user u . This per-user seed provides maximally selective one-partition routes and exposes every shared Atom copy as an explicit consolidation choice. Because the seed may exceed the memory budget, the Planner subsequently removes

Algorithm 1 Shared-Atom-Based Candidate Generation

Require: Access Atom Catalog $\{S_{A_i} \mid A_i \in \mathcal{A}\}$, users U , pruning bound d
Ensure: Seed partition plan $\mathcal{P}$, routes $\{\mathcal{P}(u)\}_{u \in U}$, inverted table I , candidate graph $\mathcal{G}$

```

1:  $(\mathcal{P}, \{\mathcal{P}(u)\}_{u \in U}, I) \leftarrow \text{PerUserSeed}(\text{catalog}, U)$ 
2:  $E_{\text{raw}} \leftarrow \emptyset$ 
3: for all  $A_i \in \mathcal{A}$  with  $|I(S_{A_i})| > 1$  do
4:    $P_c \leftarrow \text{core}(S_{A_i})$  ▷ Eq. (5)
5:   for all  $P \in I(S_{A_i}) \setminus \{P_c\}$  do
6:      $\Gamma(P_c, P) \leftarrow \Gamma(P_c, P) \cup \{S_{A_i}\}$ 
7:      $E_{\text{raw}} \leftarrow E_{\text{raw}} \cup \{(P_c, P)\}$ 
8:   end for
9: end for
10: for all  $P \in \mathcal{P}$  do
11:    $E_P \leftarrow \text{TopD}(\{e \in E_{\text{raw}} \mid P \in e\}, \phi, d)$ 
12: end for
13:  $\mathcal{G} \leftarrow (\mathcal{P}, \bigcup_{P \in \mathcal{P}} E_P)$ 
14: return  $(\mathcal{P}, \{\mathcal{P}(u)\}_{u \in U}, I, \mathcal{G})$ 

```

copies, which is called consolidation, according to their query-computation penalty. The penalty estimates how a consolidation changes query computation at the target recall; §4.4 defines this recall-aware comparison.

Only partitions sharing an Atom expose a removable copy, so SQUID avoids examining unrelated partition pairs. The Shared-Atom Graph $\mathcal{G} = (\mathcal{P}, E)$ represents these candidates, where an edge (P_x, P_y) stores shared-Atom set $\Gamma(P_x, P_y) = \mathcal{S}(P_x) \cap \mathcal{S}(P_y)$. Note that an edge identifies its shared copies but may leave the resulting partition plan to the consolidation process as described in §4.3. **An edge identifies the shared Atoms and their duplicate placements at its two endpoints; the consolidation process in §4.3 determines the resulting partition plan.**

For an Atom S_{A_i} , the Atom Inverted Table records its owner partitions as $I(S_{A_i}) = \{P \in \mathcal{P} \mid S_{A_i} \in \mathcal{S}(P)\}$. A widely replicated Atom can make $\mathcal{G}$ dense. That is, connecting all $r = |I(S_{A_i})|$ owners would create an $O(r^2)$ clique. **To optimize, SQUID selects a core-star owner and connects it to every other owner, forming a core star. The core-star owner is defined as**

$$\text{core}(S_{A_i}) = \arg \min_{P \in I(S_{A_i})} (|P|, -|S_{A_i}|/|P|). \quad (5)$$

The core star reduces the Atom’s candidate edges to $O(r)$, while omitting owner-pair candidates from the full clique. Choosing the smallest owner (i.e., the owner whose partition contains the fewest vectors) favors a compact anchor for the shared copy, while the tie-break favors the owner in which that Atom dominates the partition and is therefore less mixed with other Atoms.

When several Atoms induce the same edge, their witnesses accumulate in $\Gamma(P_x, P_y)$. SQUID ranks core-star edges with the light-weight overlap score

$$\phi(P_x, P_y) = \frac{|\Gamma(P_x, P_y)|}{\sqrt{|\mathcal{S}(P_x)| |\mathcal{S}(P_y)|}}, \quad (6)$$

The numerator counts shared Atoms, while the denominator discounts pairs whose overlap is small relative to their Atom sets. **SQUID retains each partition’s top- d raw incident edges, where $d \geq 1$, under this inexpensive pruning score and applies the full recall-aware cost model only to the retained candidates.**

Algorithm 1 summarizes this candidate-generation procedure.

Running example continued. For A_1 , the core star selects P_1 and retains (P_1, P_2) and (P_1, P_3) , while omitting (P_2, P_3) . The two

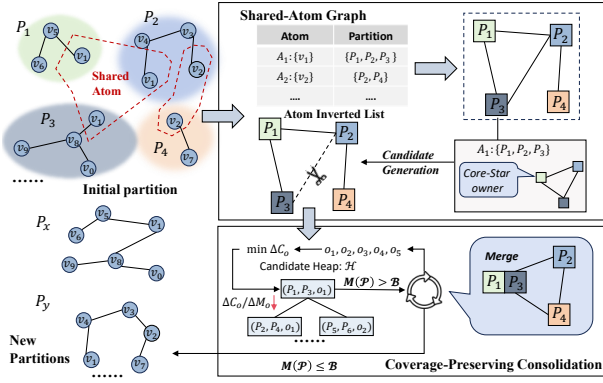


Figure 3: Shared-Atom consolidation transforms a redundant partition pair into candidate local partition plans. The upper path constructs and prunes the Shared-Atom Graph, while the lower path generates and ranks coverage-preserving consolidation alternatives.

owners of A_2 contribute (P_2, P_4) , so the graph exposes the illustrated consolidation opportunities without materializing the full owner clique.

4.3 Coverage-Preserving Consolidation

An edge locates a removable copy, but it does not determine where the surviving copy should reside. The memory saving is fixed by removing the duplicate, whereas keeping that copy with either residual side, isolating it, or reorganizing both residual sides creates different partition sizes and mixtures for the affected routes. The Planner therefore generates local alternatives and compares their query computation.

Given an edge $(P_x, P_y) \in E$, the Planner separates its endpoint partitions into the shared Atom set Ω and residual partition contents P'_x and P'_y : $\Omega = \mathcal{S}(P_x) \cap \mathcal{S}(P_y)$, $P'_x = \mathcal{S}(P_x) \setminus \Omega$, and $P'_y = \mathcal{S}(P_y) \setminus \Omega$. The shared Atom set is the source of released memory, while the residual partition contents determine the composition, and hence the selectivity, of the resulting route partitions.

Each of the following five alternatives replaces the two copies of Ω in the input partitions with one copy: absorption keeps it with one residual partition content, isolation places it in a separate partition, and reorganization either combines both residual partition contents with Ω or separates them from it:

$$\begin{aligned}
 o_1 &: \{P'_x \cup \Omega \cup P'_y\}, & o_2 &: \{P'_x \cup P'_y, \Omega\} \text{ (reorganize)} \\
 o_3 &: \{P'_x, \Omega, P'_y\}, & & \text{(isolate)} \\
 o_4 &: \{P'_x, P'_y \cup \Omega\}, & o_5 &: \{P'_x \cup \Omega, P'_y\} \text{ (absorb)}
 \end{aligned} \tag{7}$$

These alternatives exhaust the distinct ways to retain one copy of Ω while joining or separating the two residual Atom sets; empty residuals are omitted. For each affected user, the Planner replaces the two endpoint routes with output partitions covering the same Atoms and leaves unrelated route entries unchanged. The alternatives therefore preserve route coverage and authorization safety, but induce different partition sizes, mixtures, and query computation at the target recall. Let $\mathcal{P}_o$ denote the partition plan obtained by applying operation o to (P_x, P_y) . SQUID considers operations that

Algorithm 2 Coverage-Preserving Consolidation

Require: Planning $\mathcal{P}$, routes $\{\mathcal{P}(u)\}_{u \in U}$, graph $\mathcal{G}$, Atom Inverted Table $\mathcal{I}$, pruning bound d , memory budget $\mathcal{B}$, target recall τ

Ensure: Coverage-preserving partition plan and routes

```

1:  $\mathcal{H} \leftarrow \text{BuildHeap}(\mathcal{G}, \mathcal{P}, \{\mathcal{P}(u)\}_{u \in U}, \tau)$ 
2: while  $M(\mathcal{P}) > \mathcal{B}$  do
3:    $o \leftarrow \text{PopBestValid}(\mathcal{H})$ 
4:   if  $o = \perp$  then
5:      $(\mathcal{G}, \mathcal{H}) \leftarrow \text{RebuildCandidates}(\mathcal{P}, \{\mathcal{P}(u)\}_{u \in U}, \mathcal{I}, d, \tau)$   $\triangleright$  Rebuild
    from the current Atom Inverted Table
6:   else
7:     Apply  $o$  and replace the affected routes using the coverage-preserving
    rule
8:     Update  $\mathcal{P}$ ,  $\{\mathcal{P}(u)\}_{u \in U}$ ,  $\mathcal{I}$ , and the incident edges of  $\mathcal{G}$ 
9:     RefreshCandidates( $\mathcal{H}, \mathcal{G}, \mathcal{P}, \{\mathcal{P}(u)\}_{u \in U}, \mathcal{I}, d, \tau$ )
10:  end if
11: end while
12: return  $(\mathcal{P}, \{\mathcal{P}(u)\}_{u \in U})$ 

```

release memory, $\Delta M_o = M(\mathcal{P}) - M(\mathcal{P}_o) > 0$ and measures their change in query computation as $\Delta C_o = C(\mathcal{P}_o) - C(\mathcal{P})$. The Planner first prioritizes operations with non-positive ΔC_o , breaking ties by larger ΔM_o and then a fixed operation order. If more memory must be released, it orders the remaining operations by $\Delta C_o / \Delta M_o$, the computation increase per unit of saved memory. For instance, absorption can place Ω in a larger mixed partition, whereas isolation can retain a cleaner shared partition while releasing the same copy; their different route selectivities yield different ΔC_o .

Algorithm 2 summarizes the consolidation loop. The Planner repeatedly applies the best valid heap entry, updates only its affected partitions, routes, Atom Inverted Table entries, and graph edges, and rebuilds the heap when no valid entry remains. When $\mathcal{B} \geq |D|$ and $d \geq 1$, an over-budget plan contains a multiply owned Atom, so rebuilding exposes a consolidation that releases a copy and allows the loop to continue toward feasibility.

Running example continued. For (P_1, P_3) , the shared set is $\Omega = \{A_1\}$, with residual vectors $\{v_5, v_6\}$ and $\{v_9, v_8, v_0\}$. The illustrated o_1 operation combines them into a single output partition P_x . Specifically, $P_x = \{v_5, v_6, v_1, v_9, v_8, v_0\}$. This replaces two copies of A_1 with one and releases $\Delta M_{o_1} = |A_1|$. Likewise, applying o_1 to (P_2, P_4) yields $P_y = \{v_4, v_3, v_1, v_2, v_7\}$. Together, these transformations retain A_1 in P_x and P_y , consolidate A_2 into P_y , and update the affected routes while preserving coverage.

4.4 Recall-Aware Cost Model

The recall-aware cost model compares local alternatives at a common Recall@ k target. For a routed user-partition pair (u, P) , let $N = |P|$ and define its route selectivity $\rho(u, P) = \frac{|D_u \cap P|}{|P|}$, the fraction of vectors in P visible to u . Because N and $\rho(u, P)$ affect recall at a fixed HNSW [25] search depth ef_s , SQUID compares routes by the query computation required to attain the common target τ .

For route (u, P) , $x = \frac{ef_s \cdot \rho(u, P)}{k}$ is the effective visible-candidate budget per requested result. A useful planning estimator must increase with search depth and saturate as recall approaches one; to retain the same recall, a larger partition or lower route selectivity must be offset by a larger ef_s . A Hill-type saturation function [39] provides these properties by mapping the effective visible-candidate budget to a bounded recall curve with diminishing returns:

$$\widehat{\text{Rec}}(N, k, ef_s, \rho) = \frac{x^\gamma}{\lambda(N)^\gamma + x^\gamma}, \quad \lambda(N) = \lambda_0 + \lambda_1 \log(1 + N). \tag{8}$$

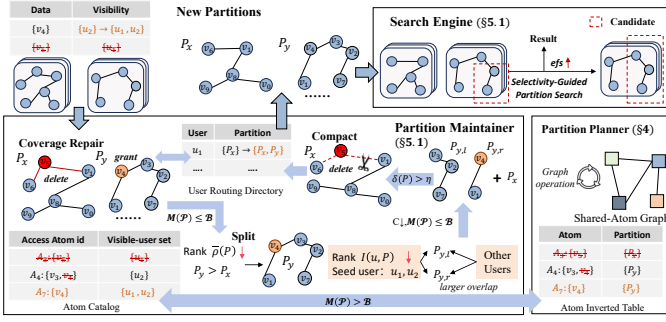


Figure 4: Runtime query processing and maintenance. The Search Engine (upper right) searches routed partitions, while the Partition Maintainer (bottom) repairs the affected partition organization after updates.

Here, $\gamma > 0$ controls how quickly recall saturates, while $\lambda(N)$ captures the logarithmic size effect; a lower route selectivity reduces x at fixed search depth. The parameters $\gamma, \lambda_0, \lambda_1$ are fitted offline, with $\lambda_1 > 0$ and $\lambda(N) > 0$ over the planned partition sizes. For fixed $N, k, \rho > 0$, and $x > 0$, $\frac{d\widehat{\text{Rec}}}{dx} = \frac{\gamma\lambda(N)^{\gamma}x^{\gamma-1}}{(\lambda(N)^{\gamma}+x^{\gamma})^2} > 0$. More precisely, $\widehat{\text{Rec}}(N, k, ef_s = 0, \rho) = 0$, and $\lim_{ef_s \rightarrow \infty} \widehat{\text{Rec}}(N, k, ef_s, \rho) = 1$. Together with monotonicity in ef_s , these limits make the model invertible at a common recall target.

Solving this monotone function gives the search depth required for a common target $\tau \in (0, 1)$:

$$\widehat{ef}_s(N, k, \tau, \rho) = \frac{k\lambda(N)}{\rho} \left(\frac{\tau}{1-\tau} \right)^{1/\gamma}. \quad (9)$$

Following prior cost-modeling practice in vector search and database systems [22, 28, 44], SQUID uses an offline-fitted latency surrogate to estimate the per-route query computation $C(u, P)$ from this depth:

$$C(u, P) = a \log(1+N) + c_0 + b \log(1+N) \cdot \widehat{ef}_s(N, k, \tau, \rho(u, P)), \quad (10)$$

where a, b , and c_0 are offline-fitted latency-surrogate coefficients; SQUID does not introduce a new latency-training procedure, but supplies the recall-normalized search depth used by this surrogate. The three terms capture graph navigation, per-partition overhead, and target-recall search effort, respectively. This model is a planning estimator for comparing candidate partition plans at a common recall target, rather than a query-level recall guarantee. The evaluation therefore validates its plan-ranking calibration separately and reports query performance at measured recall. Summing the per-route estimates over routed user-partition pairs yields $C(\mathcal{P})$, which determines ΔC_0 and the candidate-heap ordering.

5 RUNTIME QUERY PROCESSING AND MAINTENANCE

In this section, we describe how SQUID controls partition routing and search depth during search, and how it performs incremental updates to maintain partition quality under dynamic workloads.

5.1 Search over Partitions

Algorithm 3 Selectivity-Guided Partition Search

Require: $Q = (q, u, k)$ (query), D_u (visible set), $\mathcal{P}(u)$ (route), ef_0 (probe depth), dist (metric)

Ensure: R , authorized global top- k ; TopK_k retains the k closest candidates

```

1:  $\rho(u, P) \leftarrow |D_u \cap P|/|P|$ ; order  $\mathcal{P}(u)$  by decreasing  $\rho(u, P)$ 
2:  $R \leftarrow \emptyset$ ;  $\delta_k(R) \leftarrow$  its  $k$ -th distance (or  $+\infty$  if  $|R| < k$ )
3: for all  $P \in \mathcal{P}(u)$  do
4:    $C_0(P) \leftarrow \text{SEARCH}(P, q, ef_0)$ ;  $A_0(P) \leftarrow C_0(P) \cap D_u$ 
5:   if  $\exists x \in A_0(P) : \text{dist}(q, x) < \delta_k(R)$  then
6:      $\rho_0(P) \leftarrow |A_0(P)|/|C_0(P)|$ ;  $ef_P \leftarrow \lceil ef_0/\rho_0(P) \rceil$ 
7:   else if  $A_0(P) = \emptyset$  then
8:      $ef_P \leftarrow \lceil ef_0/\rho(u, P) \rceil$ 
9:   else continue
10:  end if
11:   $R \leftarrow \text{TopK}_k(R \cup (\text{SEARCH}(P, q, ef_P) \cap D_u))$ 
12: end for
13: return  $R$ 
```

Partition planning reduces replication by allowing a user route to include mixed partitions, but the resulting routes can differ substantially in the fraction of their vectors visible to the querying user. A uniform search depth would therefore waste query computation on high-selectivity routes or provide insufficient exploration for low-selectivity routes. For a query $Q = (q, u, k)$, the Search Engine obtains $\mathcal{P}(u)$ from the User Routing Directory, distributes query computation across these routed partitions, filters vectors not visible to u , and returns an approximate top- k over the valid search domain D_u .

Index setting. Each materialized partition is served by an underlying ANN index. SQUID uses HNSW [25] by default, but its routing and query-computation allocation are independent of this choice; a partition may instead use any filtered ANN index such as ACORN [27]. The chosen index generates candidates within a partition, while the Search Engine selects routes, distributes query computation, filters unauthorized candidates, and merges authorized candidates across partitions.

Search Process. Algorithm 3 summarizes the search flow. It visits routed partitions in decreasing route selectivity, uses an initial search to observe their authorized candidates, expands competitive or inconclusive partitions with a partition-specific depth, and merges the resulting authorized candidates into the global top- k result.

The algorithm adopts an established adaptive probing pattern, but operates at the visibility-route level rather than treating probing itself as a new ANNS primitive. First, graph-based filtered ANNS methods adapt candidate generation within one index [12, 27], whereas SQUID coordinates multiple partitions already selected by routing based on resolved visibility. Second, SQUID allocates partition-specific query computation using the authorized yield observed for the current query and the evolving global top- k result, rather than expanding solely according to geometric proximity or a static predicate selectivity. Third, an empty authorized probe is inconclusive, so SQUID falls back to cataloged route selectivity instead of discarding the routed partition.

It is worth noting that SQUID does not use the recall-aware cost model from §4.4 during query execution. That model compares candidate partition plans offline, whereas query execution can use current observations without evaluating and inverting the model for each partition. §A.1 details the search-depth allocation, and §A.2 formalizes how the two heuristics remain consistent in partition-planning-level expected query computation: the offline heuristic

uses its estimate to compare candidate organizations, while the online heuristic adapts partition-specific search depths within the selected organization.

5.2 Update Maintenance

Vector and resolved-visibility updates can affect three properties of the current partition organization. We call a visibility addition a *grant* and a visibility removal a *revoke*. First, a revoke or deletion must immediately prevent an ineligible vector from being returned. Second, a grant, insertion, or change in Access Atom membership may require a new Atom copy, ANN index, and route before every newly visible vector becomes reachable. Third, these physical changes may exceed the memory budget, reduce route selectivity, or leave deleted vectors physically present. SQUID therefore makes the changed visibility effective for filtering first, restores route coverage and memory feasibility in a publishable serving state, and then repairs query efficiency only within the affected region.

The Partition Maintainer performs this recovery in the background while the Search Engine continues serving queries from the latest published partitions, indexes, and routes. The Search Engine applies the latest visibility during candidate filtering, whereas a new Atom copy and route become visible to queries only after the corresponding partition and ANN index are ready.

Coverage Repair. Coverage Repair updates the Access Atom Catalog and Atom Inverted Table, records deletions in their current partitions, reconstructs changed Atom copies, and marks the affected User Routing Directory entries for route repair. A revoke operation is immediately enforced by filtering, whereas a newly created or enlarged Atom requires a searchable copy and repaired route to become reachable. Coverage Repair constructs the copy state required to cover these newly visible Atoms; the resulting copies determine whether subsequent local repair must release copies or may spend residual memory to recover selectivity.

Budget- and Selectivity-Directed Repair. Local repair first restores memory feasibility and uses only residual memory to recover query efficiency. Let $\mathcal{P}_\Delta$ denote the directly changed and newly assigned partitions, and define their one-hop repair region as

$$\mathcal{P}_\Delta^+ = \mathcal{P}_\Delta \cup \mathcal{N}_G(\mathcal{P}_\Delta), \quad (11)$$

where $\mathcal{N}_G(\mathcal{P}_\Delta)$ contains partitions connected to $\mathcal{P}_\Delta$ by retained Shared-Atom Graph edges. If $M(\mathcal{P}) > \mathcal{B}$, where $\mathcal{P}$ is the current partition plan after coverage repair, the Maintainer invokes the Partition Planner on the updated Atom and copy state within $\mathcal{P}_\Delta^+$ and processes consolidations that remove redundant copies until the memory bound is restored. Once the plan is feasible, any residual copy budget may be used by a *Split* operation, which divides a mixed partition into more selective outputs and considers only partitions in $\mathcal{P}_\Delta^+$. Partitions and indexes outside this region remain unchanged, and routes that cover no changed Atom retain their entries.

Split. Split uses residual copy budget to reduce filtering in poorly selective mixed partitions. Let $U(P) = \{u \mid P \in \mathcal{P}(u)\}$ denote the users routed to P . To select a partition, SQUID considers mixed partitions $P \in \mathcal{P}_\Delta^+$ with $|U(P)| \geq 2$ in increasing average route selectivity $\bar{\rho}(P) = \frac{1}{|U(P)|} \sum_{u \in U(P)} \rho(u, P)$, breaking ties by lower $\min_{u \in U(P)} \rho(u, P)$ and then larger $|P|$. Within the selected partition,

the filtering impact $I(u, P) = |D_u \cap P|(1 - \rho(u, P))$ prioritizes users whose queries discard the most candidates. Among these users, SQUID chooses as seeds the pair minimizing the vector-weighted overlap $\omega_P(u, v) = \frac{|P \cap D_u \cap D_v|}{|P \cap (D_u \cup D_v)|}$, thereby separating users with the least overlapping visible vectors. Each remaining user, processed in decreasing $I(u, P)$, joins the seed group with which it shares more vectors through common visible Atoms; ties favor the group with the smaller total filtering impact. An Atom visible to users in both groups is replicated into both output partitions, whereas every other Atom is placed only in the output for the group containing users who can see it. Replacing P with these outputs and repairing the affected routes yields a coverage-preserving candidate plan $\mathcal{P}'$. SQUID accepts the split only if $C(\mathcal{P}') < C(\mathcal{P})$ and $M(\mathcal{P}') \leq \mathcal{B}$.

Route Repair. The accepted Atom copies determine the final route repair. For each affected user, the Maintainer retains a complete route when possible; otherwise, it greedily adds the partition covering the most uncovered Atoms, preferring higher route selectivity and smaller size on ties. The repaired routes satisfy $D_u \subseteq \bigcup_{P \in \mathcal{P}(u)} P$ for every $u \in U$.

Compaction and Batch Completion. After the partition plan and repaired routes are fixed, the Maintainer reclaims physically retained deleted vectors through compaction without changing Atom copies or user routes. For partition P , let $d(P)$ count vectors recorded for deletion but still physically present and let $|P|$ count its valid vectors, giving $\delta(P) = \frac{d(P)}{|P| + d(P)}$. SQUID compacts P when $\delta(P) \geq \eta$, where η is a configurable compaction threshold [28]. The Maintainer then materializes every new, changed, or compacted partition from its valid vectors and builds its ANN index before publishing the corresponding routes and metadata. Consequently, revokes take effect immediately, while newly visible vectors become reachable after their background-built copies, indexes, and routes are published. The batch completes after the compacted state is saved, while unrelated partitions, indexes, and routes remain available without reconstruction.

Running Example. Figure 4 applies a batch that deletes v_5 and grants u_1 access to v_4 . The Search Engine immediately excludes v_5 under the changed visibility, while Coverage Repair records its deletion and places the new Atom containing v_4 in P_y . Because the resulting plan remains within the memory budget, SQUID skips consolidation and considers splitting P_y only if the split reduces $C(\mathcal{P})$ within the residual budget. Route repair then extends u_1 's route to P_y , and v_4 becomes reachable after the corresponding index and route are published. Finally, the Maintainer removes v_5 from P_x through compaction when $\delta(P_x) \geq \eta$, after which the batch completes.

6 EVALUATION

Our evaluation answers four research questions:

RQ1. Authorization-policy independence. Can SQUID construct a partition plan from resolved visibility and query its materialized partition organization without relying on the policy representation that produced it? (§6.2.3)

RQ2. Memory-performance trade-off. Within a fixed memory budget, how does SQUID trade memory consumption for query performance relative to shared, fully isolated, and budget-constrained partition organizations? (§6.2.1, §6.2.2 & §6.3)

Table 2: Datasets and visibility workloads. Avg. fan-out: authorized users per vector; ROLE/RLS: full-ROLE footprint relative to one-copy RLS.

Dataset	Dim.	Size	Visibility Workload	Avg. Fan-out	Avg. Sel.	ROLE/RLS
SIFT [23]	128	1M	Tree-based [36, 44]	36.1	3.61%	$\sim 3.5\times$
Wiki [35]	768	485K	ERBAC [21, 44]	69.4	6.94%	$\sim 6.2\times$
YFCC [34]	192	1M	YFCC-ABAC [14, 30]	13.4	1.34%	$\sim 25.7\times$

RQ3. Planning and maintenance. How much time does SQUID require to construct a partition plan and maintain its materialized partition organization, and how does QPS change across update batches? (§6.4)

RQ4. Mechanisms and robustness. How much do the Search Engine and recall-aware cost model contribute, how sensitive are planning and query performance to the main parameters, and does SQUID remain compatible with a filtered ANN index? (§6.5)

6.1 Experimental Setup

We evaluate both partition planning and execution of the resulting partition organization in a real database system. The planning experiments measure the replication budget, materialized database footprint, planning time, and maintenance time. The query experiments measure Recall@ k and end-to-end throughput after routing, ANN search, authorization filtering, and result merging. This separation lets us distinguish a partition plan that is inexpensive to construct from one whose materialized partition organization is also effective to query.

Platform. All experiments run in a Docker container allocated 16 CPU cores and 256 GB of main memory. The container runs PostgreSQL 17 [28] with pgvector 0.8.0. Unless stated otherwise, all baselines use HNSW [25] as their ANN index. We use the same database configuration, query sequence, and warm-up procedure for every comparison.

Datasets and visibility workloads. Table 2 summarizes SIFT [23], Wiki [35], and YFCC [34], together with the following resolved-visibility workloads paired with them.

- **ERBAC** [21, 44] uses 40 functional and 100 business roles; each business role aggregates at most three functional roles, and each user receives at most three business roles.
- **Tree-based** [36, 44] uses 100 roles in a height-4 hierarchy with 3–4 children per internal role, producing nested visible-vector sets.
- **YFCC-ABAC** [14, 30] uses 50 two-label attribute rules derived from YFCC metadata, with two rules per user, overlap probability 0.5, minimum support 16, and 256 candidate clauses per rule.

Baselines. We compare SQUID with four baseline categories:

- **RLS** uses one shared HNSW index and enforces resolved visibility through PostgreSQL Row-Level Security during query execution.
- **ROLE Partition** materializes a role-local search space and serves as an unconstrained isolation reference when its replication exceeds the budget.
- **HQI** [26], **HONEYBEE** [44], **VEDA** [13], and **EFFVEDA** [13] are policy-aware partitioning methods that trade replication for lower query computation.

- **ACORN** [27] is a filtered ANN index used alone and inside SQUID’s mixed partitions.

ERBAC and Tree-based provide native roles; hence, they can pair with the datasets directly. For YFCC-ABAC, we represent each user as a virtual role with the same visible-vector set, so role-oriented baselines receive 1,000 virtual roles.

Queries and metrics. We use the exact visible top- k results within D_u as ground truth and evaluate every method on the same query set; $k = 10$ unless stated otherwise. Each QPS point reports the mean over five repetitions of 1,000 sampled queries after one warm-up round. QPS is the end-to-end throughput measured through PostgreSQL, whereas the query-time experiments measure the execution time of one SQL query to isolate the cost of routing and search over the materialized partition organization from client-side concurrency. We sweep search depth to obtain QPS-Recall@ k curves and use the smallest depth that reaches each recall target.

Common protocol and parameter tuning. We express the bounded memory budget in planning as a replication ratio: the number of materialized vector copies after partitioning divided by the number of input vectors; a $1.5\times$ budget therefore permits at most 1.5 copies per input vector on average. The realized database footprint, including physical vector relations, per-partition ANN indexes, and planning metadata, is measured separately in §6.3.

All methods use the same resolved visibility, query set, ground truth, database configuration, query sequence, and warm-up procedure. We build SQUID and our implementations of VEDA and EFFVEDA on top of the released HONEYBEE codebase, and reuse its implementations of RLS, ROLE Partition, HQI, and ACORN. HNSW-based methods share the distance metric L_2 and construction parameters $M = 16$ and $ef_{\text{construction}} = 64$. For each fixed partition organization, we report the smallest value reaching the target recall. A method that cannot construct a partition plan within the requested replication budget or the three-hour construction limit is omitted and is not queried in our results.

6.2 Memory-Efficiency Trade-offs

Figure 5 reports end-to-end QPS and single-query execution time for five dataset–workload pairs under the common $1.5\times$ vector-copy budget. ROLE, as the upper bound, represents unconstrained full isolation, while RLS represents one shared index as a filtered-index design. The remaining methods are evaluated under the common budget. Each curve sweeps search depth for a fixed partition plan, and a missing curve indicates that the method did not produce a budget-feasible plan within three-hour partition construction.

6.2.1 Budget-Constrained Query Performance. Across the dataset–workload pairs in Figure 5, SQUID forms the strongest high-recall QPS and query-time curves among methods constrained to $1.5\times$ vector copies. Near 0.95 recall on SIFT+ERBAC, SQUID provides $1.29\times$ the QPS of EFFVEDA and $2.05\times$ that of HONEYBEE. On Wiki+Tree-based, it improves QPS by 13% over HONEYBEE, 25% over EFFVEDA, and more than $8\times$ over RLS. On SIFT+Tree-based, SQUID reduces matched-recall query time by 24%–37% relative to HONEYBEE, VEDA, and EFFVEDA and by approximately $7\times$ relative to RLS. On YFCC+ABAC, it reaches $4.7\times$ EFFVEDA’s QPS and reduces its query time by $3.8\times$. These results show that SQUID’s

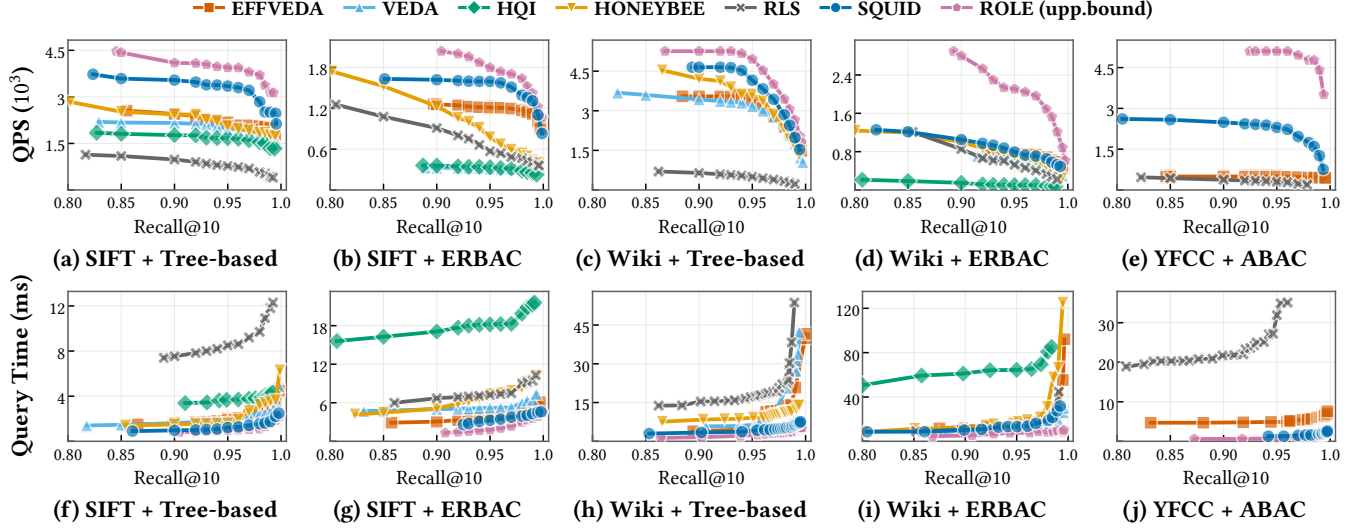


Figure 5: QPS–Recall@10 (top) and query-time–Recall@10 (bottom) across five dataset–workload pairs. Constrained methods use 1.5 $\times$ copies; ROLE is unconstrained and fully materialized.

Table 3 High-recall QPS and memory on SIFT+Tree-based; memory ratios are normalized to SQUID.

Method	#Partitions	Mem.(MB)	Mem.($\times$)	QPS
HONEYBEE	77	4236.000	2.002	2752.000
EFFVEDA	100	3968.555	1.875	3240.990
VEDA	97	3933.430	1.859	3355.691
SQUID	58	2116.008	1.000	3545.451

Table 4 Candidate-degree sensitivity (default in bold).

Top- d	Query Time (ms)	Planning Time (s)
4	3.663	2.020
8	4.945	2.188
16	3.149	2.064
32	4.163	2.083
64	3.468	2.157

advantage persists across image and text vectors and across role- and attribute-derived visibility.

6.2.2 Cost of Full Isolation. ROLE bounds the performance gap to full isolation but exceeds the 1.5 $\times$ budget. Under Tree-based visibility, SQUID uses 57% fewer vector copies and remains within 1.26 $\times$ of ROLE’s latency at 0.95 recall. Under ERBAC, SQUID uses 75% fewer copies and remains within 1.84 $\times$ of ROLE’s matched-recall latency. Closing the remaining latency gap therefore requires ROLE to store 2.3 $\times$ –4.0 $\times$ as many vector copies as SQUID.

6.2.3 Authorization-policy Independence. Tree-based and ERBAC derive visibility from role structures, whereas YFCC-ABAC derives it from attribute rules. SQUID consumes the same resolved-visibility representation and runs the same Planner without policy-specific configuration in all three cases. For fairness, policy-oriented baselines receive native roles for Tree-based and ERBAC and 1,000 visibility-equivalent virtual roles for YFCC-ABAC. On YFCC-ABAC, HONEYBEE, VEDA, and HQI exceed the construction limit, while EFFVEDA and SQUID finish and SQUID attains the strongest measured high-recall performance. The result establishes policy independence for the tested role- and attribute-derived workloads, without claiming coverage of every authorization policy.

6.3 Memory Footprint

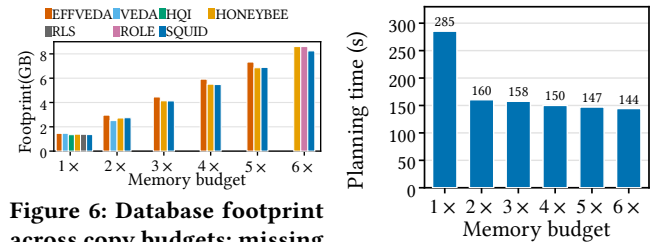


Figure 6: Database footprint across copy budgets; missing bars denote unavailable settings.

Table 3 compares realized memory and QPS at measured Recall@10 near 0.95 on SIFT+Tree-based. SQUID uses a 1.5 $\times$ vector-copy budget, whereas the compared baselines are not constrained to this budget. SQUID reaches 3,545 QPS with 2,116 MB, the smallest footprint and highest QPS among the compared partitioning methods. HONEYBEE, EFFVEDA, and VEDA consume 2.00 $\times$, 1.88 $\times$, and 1.86 $\times$ SQUID’s memory while providing 0.78 $\times$, 0.91 $\times$, and 0.95 $\times$ its QPS, respectively.

Figure 6 further reports realized footprint across vector-copy budgets. Because the planning budget excludes per-partition indexes and method-specific metadata, the footprint includes materialized vector relations, their ANN indexes, and persistent planning metadata but excludes common source tables and transient construction memory. The 1 $\times$ reference is the 1.385 GB footprint of one indexed SIFT copy. The differences at equal copy budgets confirm that vector-copy count and realized database memory must be reported separately.

6.4 Planning and Maintenance

6.4.1 Partition Planning. We first measure the Partition Planner’s initialization time on SIFT+ERBAC. Figure 7 reports the time from resolved visibility to a completed partition plan across 1 $\times$ –6 $\times$ vector-copy budgets, excluding partition materialization and ANN-index construction. Planning takes 285.4 seconds at 1 $\times$ and 144.4

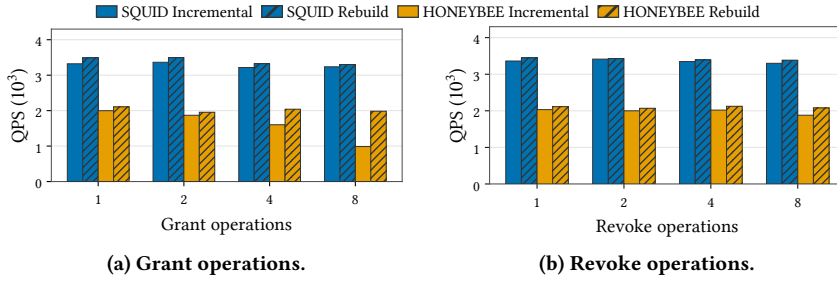


Figure 8: Maintenance QPS after grant/revoke batches.

seconds at 6 $\times$, and every tested budget completes within five minutes. The Planner therefore produces every tested organization for this workload within the evaluation’s construction limit.

6.4.2 Incremental Maintenance. We evaluate whether incremental maintenance preserves **query efficiency** relative to complete rebuilding. On SIFT+Tree-based, each batch contains $n \in \{1, 2, 4, 8\}$ grant or revoke operations. For each system, *incremental* updates the existing partition organization, whereas *rebuild* reconstructs it from the same post-update visibility. Both modes use the same query workload and recall target, so their QPS difference isolates the query-performance loss caused by incremental maintenance. Figure 8 reports the grant and revoke results separately.

For grants, SQUID’s incremental QPS remains within 4.75% of complete rebuilding across all batch sizes. For revokes, the corresponding gap remains within 2.53%. Incremental maintenance sustains 3.2–3.4K QPS, and the gap from rebuilding does not widen as the number of operations increases. Affected-region repair therefore preserves most of the query efficiency recovered by complete reconstruction. HONEYBEE remains close to rebuilding only for small batches. At eight grants, its incremental QPS falls 50.25% below rebuilding, while at eight revokes it falls 9.51% below rebuilding. SQUID’s incrementally maintained organization remains 1.58–1.72 $\times$ faster than fully rebuilt HONEYBEE in every tested setting. Grants are more disruptive because they may require new route coverage and placements, whereas revokes primarily trigger cleanup. SQUID handles both update classes within the affected region without accumulating substantial query-performance loss.

Figure 9 evaluates **recall recovery** while foreground queries execute continuously. We replay a fixed panel of 50 update-affected queries every 10 seconds and report their mean Recall@ k . Three batches of 100 updates begin at approximately 5, 85, and 165 seconds. All methods use HNSW with method-specific search parameters tuned to their steady-state recall.

SQUID starts near 0.975 recall and falls to approximately 0.900, 0.930, and 0.935 after the three update batches. After each drop, recall returns to 0.965–0.975 as SQUID repairs and publishes the affected routes while structural maintenance continues in the background. EFFVEDA initially falls to approximately 0.86, and its later recovery coincides with publication of a completed background reconstruction. HONEYBEE does not complete its first reconstruction within the 240-second observation window. Its successive update effects therefore accumulate, and recall declines to approximately 0.82 after the third batch. These results show that rapid route repair

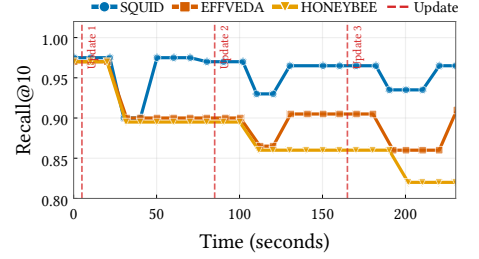


Figure 9: Recall during incremental maintenance.

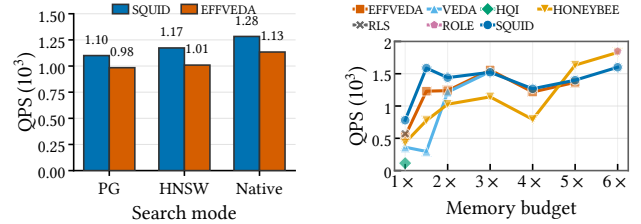


Figure 10: Search Engine execution modes.

Figure 11: QPS at Recall@10 $\geq$ 95% across copy budgets.

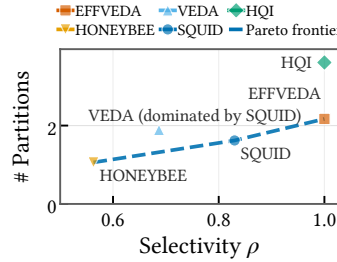


Figure 12: Route-quality Pareto frontier on SIFT+Tree-based. Higher s and lower p are preferable. The dashed line connects the observed Pareto-optimal methods; it is not a fitted frontier.

limits SQUID’s recall-degradation window, whereas reconstruction-based maintenance can suffer persistent loss when rebuilding cannot keep pace with updates.

6.5 System Mechanisms and Sensitivity

6.5.1 Partition-Plan Quality. We analyze the route-quality measurements of the partition organizations produced on SIFT+Tree-based under an approximately 1.5 $\times$ vector-copy setting. We summarize each organization by mean routing selectivity s , obtained by averaging $\rho(u, P)$ over routed user-partition pairs, and mean route fan-out p , the average number of partitions on a user route. Higher s indicates less filtering within routed partitions, whereas lower p avoids distributing a query across additional ANN indexes. Figure 12 visualizes the observed Pareto frontier in the (s, p) plane, while Table 5 reports the exact route-quality values and realized footprints.

HONEYBEE has the lowest fan-out among the plotted methods, $p = 1.06$, but also the lowest routing selectivity, $s = 0.563$. SQUID increases selectivity to 0.830, a 47% improvement, while increasing fan-out by 0.56 partitions per user to 1.62. Table 5 shows that

Table 5: Partition-organization measurements; footprint in GB.

Method	s	p	GB
HONEYBEE	0.563	1.06	2.117
SQUID	0.830	1.62	2.115
VEDA	0.687	1.89	2.241
EFFVEDA	1.000	2.17	2.343
HQI	1.000	3.61	1.389

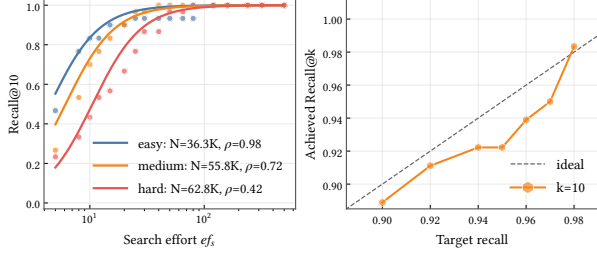


Figure 13: Recall-model fit (a) and achieved recall at the predicted search depth (b).

this improvement does not require a larger footprint: SQUID and HONEYBEE consume essentially the same space (2.115 versus 2.117 GB). VEDA reaches only $s = 0.687$ with a higher fan-out of 1.89, so SQUID improves both plotted properties while also using a smaller footprint (2.115 versus 2.241 GB).

EFFVEDA and HQI both obtain pure routes with $s = 1.000$, but their average fan-outs increase to 2.17 and 3.61, respectively. Relative to SQUID, HQI increases selectivity from 0.830 to 1.000 and reduces footprint from 2.115 to 1.389 GB, but requires 1.99 additional routed partitions per user. Their comparison therefore represents a trade-off rather than a dominance relation. Overall, HONEYBEE, SQUID, and EFFVEDA lie on the observed Pareto frontier in the (s, p) plane, representing progressively higher routing selectivity at the cost of greater route fan-out.

6.5.2 Query-Execution Benefit. We isolate query execution under a $1.5\times$ budget on SIFT+ERBAC by holding each method’s plan and query stream fixed. Figure 10 compares PostgreSQL HNSW compensation, direct HNSW execution, and each method’s native execution mode. SQUID reaches 1,283 QPS, exceeding direct HNSW by 9.4%, PostgreSQL compensation by 16.6%, and EFFVEDA’s native mode by 13.1%. This ablation measures the aggregate benefit of SQUID’s execution path but does not separately attribute the gain to route ordering and adaptive effort allocation.

6.5.3 Recall-Aware Cost Calibration. We fit the recall model by weighted nonlinear least squares over measured recall, partition size, route selectivity, k , and ef_s . Figure 13 compares fitted and measured Recall@10 curves and evaluates the search depths predicted for target recall. The fitted curves reproduce the measured saturation trend and preserve the ordering of the three route settings. Achieved recall differs from the target by at most approximately 0.03, although the model undershoots between 0.92 and 0.97. The model is therefore suitable for approximate plan ranking but not as a recall guarantee, so all query results use measured recall after independent search-depth sweeps.

6.5.4 Candidate-Graph Sparsity. We vary $d \in \{4, 8, 16, 32, 64\}$ on SIFT+Tree-based while holding the workload and memory setting fixed. Table 4 shows planning-stage times of 2.020–2.188 seconds and query times of 3.15–4.95 ms. The lowest degree $d = 4$ reaches 3.663 ms with the shortest planning time, but query time is non-monotonic and does not identify a query-optimal degree. We use $d = 32$ to retain more candidates at only 0.063 seconds, or 3.1%, more planning time than $d = 4$. The bounded ranges indicate limited sensitivity over the tested degrees.

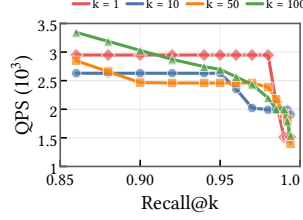


Figure 14: QPS-Recall@k across result sizes.

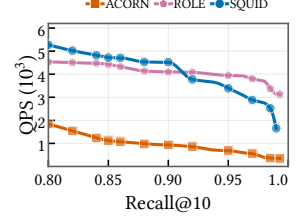


Figure 15: ACORN compatibility on SIFT+Tree-based at $1.2\times$ copies.

6.5.5 Memory-Budget Sensitivity. We plan SIFT+ERBAC from $1\times$ to $6\times$ and select the smallest search depth reaching Recall@10 near 0.95 for each method and budget. Figure 11 shows that SQUID produces a plan at every budget, with QPS ranging from 780 to 1,600. At $1.5\times$, SQUID reaches 1,588 QPS, 29% above EFFVEDA and $2.05\times$ HONEYBEE. SQUID does not dominate at the largest budgets, where HONEYBEE and unconstrained ROLE reach higher QPS. Its advantage is therefore concentrated under tight memory budgets.

6.5.6 Result-Size Sensitivity. We reuse one partition plan for $k \in \{1, 10, 50, 100\}$ and independently sweep search depth for each result size. Figure 14 reports the resulting QPS-Recall@k curves. Near 0.95 recall, $k=1$ reaches approximately 3.1K QPS, while $k=10, 50$, and 100 reach 2.5–2.9K QPS. The limited spread among the multi-result settings shows that high-recall throughput is moderately sensitive to k over the tested range.

6.5.7 Compatibility with Filtered ANN Indexes. We test whether SQUID remains useful when its mixed partitions use ACORN rather than HNSW. On SIFT+Tree-based, SQUID uses a $1.2\times$ copy budget, ACORN for mixed partitions, and HNSW for pure partitions without refitting its HNSW-derived cost model. Figure 15 compares SQUID+ACORN with one collection-wide ACORN index and fully materialized ROLE partitions. SQUID+ACORN exceeds collection-wide ACORN throughout their common recall range. Near 0.95 recall, SQUID+ACORN reaches about 3.0K QPS, roughly $6\times$ collection-wide ACORN’s throughput; the ratio remains about $5\times$ near 0.98 recall. ROLE is faster over part of the high-recall range but requires roughly $3.5\times$ vector copies, compared with SQUID+ACORN’s $1.2\times$ budget. This single-workload result establishes compatibility rather than index-independent optimality: partitioning remains beneficial when ACORN replaces HNSW in mixed partitions, even though the recall-aware cost model is not refitted for ACORN.

7 RELATED WORK

General ANN indexing and systems. ANN search has been studied through locality-sensitive hashing [2, 6], vector quantization [4, 11], and graph navigation [24, 25]. Database systems such as PASE [42] and mixed vector-relational access paths [29] integrate vector search with relational execution. These methods optimize search within a fixed physical collection, whereas SQUID determines visibility-derived search spaces and can use an ANN index inside each partition.

Filtered-index methods. Filtered ANNS systems attach predicates to a shared index rather than materializing a physical layout for each query scope [12, 27, 38, 41, 45]. ACORN [27] and

Filtered-DiskANN [12] modify graph traversal, while SeRF [45] and SIEVE [22] build or select predicate-aware structures. HQANN [40] similarly combines vector search with structured predicates. These methods remain effective substrates for SQUID, but a collection-wide filtered index can still examine many invisible vectors when a user’s resolved visible set is sparse.

Partition-based methods. Partition-based systems expose searchable physical units and route queries only to units that may contain valid results [13, 16, 26, 44]. HQI [26], Curator [16], HONEYBEE [44], and VEDA [13] exploit roles, groups, labels, or visible-set relations and use replication to trade memory for lower query computation. Their planning units are commonly derived from a stable authorization structure, so policy evolution may require reconstruction or produce coarse mixed partitions. SQUID instead derives Access Atoms from resolved visibility and jointly decides copy retention, common-recall route cost, and local repair without assuming one policy representation.

8 CONCLUSION

Access-controlled vector search requires physical organization over resolved visibility rather than only query-time filtering or partitions derived from a particular authorization policy. SQUID represents visibility-equivalent vectors as Access Atoms and uses sparse Shared-Atom candidate generation with recall-aware ranking to retain beneficial copies under a memory budget. Its route-aware Search Engine coordinates search over the resulting mixed partitions, while the Partition Maintainer enforces current visibility and repairs affected copies, routes, indexes, and metadata without routine global reconstruction. Across RBAC- and ABAC-derived workloads, SQUID improves throughput at matched recall under constrained replication, remains compatible with ACORN, and localizes maintenance after vector and visibility updates.

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A ALIGNING PLANNING AND QUERY EXECUTION

SQUID uses different heuristics for offline partition planning and query execution because the two stages operate under different latency and information constraints. Before indexes are materialized, the Partition Planner must compare many candidate partition plans without executing the future query workload for each plan. It therefore uses the recall-aware cost model in §4.4 as an offline surrogate for expected query computation at a common recall target. At query time, applying and inverting this model for every routed partition would add computation to the latency-critical path, and its static inputs cannot capture the candidate distribution of the current query. The Search Engine therefore uses the selectivity-guided probing policy in §5.1 to allocate search depth from query-local observations.

A.1 Detailed Search Procedure

For a query $Q = (q, u, k)$, the User Routing Directory returns the routed partitions $\mathcal{P}(u)$ and their route selectivities $\rho(u, P) = |D_u \cap P|/|P|$. The Search Engine visits these partitions in decreasing route selectivity so that high-selectivity routes can establish a competitive global result early. It maintains an authorized candidate set R , where $\delta_k(R)$ is the distance of the current k -th candidate if $|R| \geq k$ and $+\infty$ otherwise. For each routed partition P , an initial search with depth ef_0 returns candidates $C_0(P)$ and the authorized subset

$$A_0(P) = \{x \in C_0(P) \mid x \in D_u\}.$$

If $A_0(P)$ contains a candidate closer than $\delta_k(R)$, SQUID estimates the query-local authorized ratio

$$\rho_0(P) = \frac{|A_0(P)|}{|C_0(P)|}$$

and expands the search to

$$ef_P = \left\lceil \frac{ef_0}{\rho_0(P)} \right\rceil.$$

If $A_0(P) \neq \emptyset$ but none of its candidates can improve the current result, SQUID does not expand that partition. An empty $A_0(P)$, however, is inconclusive because the initial search may miss sparse authorized vectors. SQUID therefore falls back to the cataloged route selectivity and uses

$$ef_P = \left\lceil \frac{ef_0}{\rho(u, P)} \right\rceil.$$

After each expansion, the Search Engine filters unauthorized candidates, merges authorized candidates into R , and updates $\delta_k(R)$. The tighter threshold makes later expansion decisions depend on the results already obtained from earlier routed partitions. Algorithm 3 gives the corresponding control flow.

A.2 Heuristic Alignment

Using two heuristics raises a compositional question. The Partition Planner selects a partition plan using its offline estimate; after materialization, the resulting partition organization is served by the Search Engine’s selectivity-guided probing policy. The relevant requirement is therefore not that the two heuristics produce identical search depths. Instead, the offline estimate should preserve the

ordering of the expected query computation that the selectivity-guided probing policy achieves on candidate partition plans. This section gives sufficient conditions under which the two-stage design satisfies this requirement.

Scope. Fix a workload distribution, target recall τ , and selectivity-guided probing policy π . For a feasible partition plan $\mathcal{P}$, let $C^\pi(\mathcal{P})$ denote the expected query computation obtained after materializing the plan and having the Search Engine execute $\mathcal{P}$ using π under this setting. Let $\widehat{C}(\mathcal{P})$ denote the offline planning estimate, corresponding to $C(\mathcal{P})$ in §4.4. The analysis assumes a bounded difference between $\widehat{C}(\mathcal{P})$ and $C^\pi(\mathcal{P})$ over the candidate partition plans being compared. It establishes the resulting composition and ordering guarantees, but it does not establish that the fitted model has such a uniform bound for every workload or recall target. It also does not guarantee the recall of an individual query or directly order wall-clock latency.

At one step of the Partition Planner, let $\mathfrak{C}$ be the finite set of candidate partition plans produced by the currently valid local consolidation operations. Define the candidate selected by the recall-aware cost model and the best candidate under the selectivity-guided probing policy as

$$\widehat{\mathcal{P}} \in \arg \min_{\mathcal{P} \in \mathfrak{C}} \widehat{C}(\mathcal{P}), \quad \mathcal{P}_\pi^\star \in \arg \min_{\mathcal{P} \in \mathfrak{C}} C^\pi(\mathcal{P}).$$

PROPOSITION A.1 (COMPOSITION OF PARTITION PLANNING AND QUERY EXECUTION). *Suppose that*

$$\left| \widehat{C}(\mathcal{P}) - C^\pi(\mathcal{P}) \right| \leq \epsilon$$

for every $\mathcal{P} \in \mathfrak{C}$. Then

$$C^\pi(\widehat{\mathcal{P}}) \leq C^\pi(\mathcal{P}_\pi^\star) + 2\epsilon.$$

PROOF. The error bound and the definition of $\widehat{\mathcal{P}}$ give

$$C^\pi(\widehat{\mathcal{P}}) \leq \widehat{C}(\widehat{\mathcal{P}}) + \epsilon \leq \widehat{C}(\mathcal{P}_\pi^\star) + \epsilon \leq C^\pi(\mathcal{P}_\pi^\star) + 2\epsilon.$$

□

Proposition A.1 formalizes why the two heuristics can be used together. The Partition Planner need not reproduce the per-query decisions of π ; it only needs to estimate their expected consequence well enough to select the partition plan. Under the stated error bound, the candidate selected by the recall-aware cost model incurs at most 2ϵ more expected query computation than the best candidate available at that planning step when both use the same selectivity-guided probing policy. We next give pairwise and consolidation-operation-level forms of this result.

PROPOSITION A.2 (ROBUST PARTITION-ORGANIZATION ORDERING). *Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be two feasible partition plans. Suppose that the planning estimator has absolute error at most ϵ on both candidates:*

$$\left| \widehat{C}(\mathcal{P}_i) - C^\pi(\mathcal{P}_i) \right| \leq \epsilon, \quad i \in \{1, 2\}.$$

If

$$\widehat{C}(\mathcal{P}_1) + 2\epsilon < \widehat{C}(\mathcal{P}_2),$$

then

$$C^\pi(\mathcal{P}_1) < C^\pi(\mathcal{P}_2).$$

PROOF. The assumed error bound gives

$$C^\pi(\mathcal{P}_1) \leq \widehat{C}(\mathcal{P}_1) + \epsilon$$

and

$$C^\pi(\mathcal{P}_2) \geq \widehat{C}(\mathcal{P}_2) - \epsilon.$$

The separation condition implies

$$\widehat{C}(\mathcal{P}_1) + \epsilon < \widehat{C}(\mathcal{P}_2) - \epsilon.$$

Combining these inequalities yields

$$C^\pi(\mathcal{P}_1) \leq \widehat{C}(\mathcal{P}_1) + \epsilon < \widehat{C}(\mathcal{P}_2) - \epsilon \leq C^\pi(\mathcal{P}_2).$$

Therefore, $C^\pi(\mathcal{P}_1) < C^\pi(\mathcal{P}_2)$. $\square$

Proposition A.2 shows that exact prediction is unnecessary for comparing partition plans. An estimation error can reverse the ordering only when the estimated costs of two candidates differ by at most 2ϵ . Consequently, decisions with a sufficiently large estimated cost margin remain robust even when the model does not accurately predict the search depth required by every individual query.

The Partition Planner applies this reasoning to local consolidation operations. For an operation o that changes the current partition plan $\mathcal{P}$ into $\mathcal{P}_o$, define the executed and estimated changes in query computation as

$$\Delta C_o^\pi = C^\pi(\mathcal{P}_o) - C^\pi(\mathcal{P})$$

and

$$\widehat{\Delta C}_o = \widehat{C}(\mathcal{P}_o) - \widehat{C}(\mathcal{P}).$$

COROLLARY A.3 (ROBUST CONSOLIDATION-OPERATION CLASSIFICATION). *Suppose that*

$$\left| \widehat{C}(\mathcal{P}') - C^\pi(\mathcal{P}') \right| \leq \epsilon$$

for the current partition plan $\mathcal{P}$ and every candidate partition plan $\mathcal{P}'$. Then

$$\left| \widehat{\Delta C}_o - \Delta C_o^\pi \right| \leq 2\epsilon.$$

In particular, if $\widehat{\Delta C}_o < -2\epsilon$, then $\Delta C_o^\pi < 0$, and operation o reduces the expected query computation under π . Likewise, if $\widehat{\Delta C}_o > 2\epsilon$, then $\Delta C_o^\pi > 0$.

PROOF. By the triangle inequality,

$$\begin{aligned} \left| \widehat{\Delta C}_o - \Delta C_o^\pi \right| &= \left| \widehat{C}(\mathcal{P}_o) - C^\pi(\mathcal{P}_o) - \left(\widehat{C}(\mathcal{P}) - C^\pi(\mathcal{P}) \right) \right| \\ &\leq \left| \widehat{C}(\mathcal{P}_o) - C^\pi(\mathcal{P}_o) \right| + \left| \widehat{C}(\mathcal{P}) - C^\pi(\mathcal{P}) \right| \\ &\leq 2\epsilon. \end{aligned}$$

If $\widehat{\Delta C}_o < -2\epsilon$, then

$$\Delta C_o^\pi \leq \widehat{\Delta C}_o + 2\epsilon < 0.$$

The case $\widehat{\Delta C}_o > 2\epsilon$ follows symmetrically. $\square$

When multiple memory-reducing operations remain, SQUID ranks them by the estimated increase in query computation per unit of saved memory. Let $\Delta M_o > 0$ denote the memory released by operation o , and define

$$r_o^\pi = \frac{\Delta C_o^\pi}{\Delta M_o}, \quad \widehat{r}_o = \frac{\widehat{\Delta C}_o}{\Delta M_o}.$$

Memory consumption is computed directly from the materialized copies, so ΔM_o is not estimated by the recall-aware cost model.

COROLLARY A.4 (ROBUST ORDERING OF MEMORY-RELEASE OPERATIONS). *For two candidate operations o_1 and o_2 , suppose the query-computation estimation error is bounded by ϵ for the current and resulting partition plans. If*

$$\widehat{r}_{o_1} + \frac{2\epsilon}{\Delta M_{o_1}} + \frac{2\epsilon}{\Delta M_{o_2}} < \widehat{r}_{o_2},$$

then

$$r_{o_1}^\pi < r_{o_2}^\pi.$$

PROOF. Corollary A.3 implies

$$\left| \widehat{r}_o - r_o^\pi \right| = \frac{\left| \widehat{\Delta C}_o - \Delta C_o^\pi \right|}{\Delta M_o} \leq \frac{2\epsilon}{\Delta M_o}.$$

Therefore,

$$r_{o_1}^\pi \leq \widehat{r}_{o_1} + \frac{2\epsilon}{\Delta M_{o_1}} < \widehat{r}_{o_2} - \frac{2\epsilon}{\Delta M_{o_2}} \leq r_{o_2}^\pi.$$

Hence, $r_{o_1}^\pi < r_{o_2}^\pi$. $\square$

Interpretation and limitations. The results above explain why exact cost prediction is unnecessary for robust planning decisions, but they remain conditional on the assumed error bound. They do not show that the fitted recall-aware cost model has a uniform ϵ for every workload, partition plan, or recall target. The bound may be particularly loose at high recall targets, where a small recall-prediction error can induce a large error after the saturation curve is inverted. SQUID consequently uses the recall-aware cost model to compare expected query computation during offline partition planning rather than to provide a query-time recall guarantee.

The definition of C^π fixes the workload distribution, recall target, and selectivity-guided probing policy. The propositions concern query computation and do not directly guarantee wall-clock latency. The composition guarantee is local to each candidate-selection step and does not turn SQUID's greedy consolidation sequence into a globally optimal solution of MBP². They also leave candidates inside the uncertainty margin unordered: when the estimated query computation of two partition plans differ by at most 2ϵ , either ordering may be consistent with the assumed error bound. In this regime, SQUID may retain the current partition plan or apply a deterministic secondary criterion, such as greater memory reduction, instead of treating an uncertain cost difference as conclusive.

B RECALL-MODEL TRAINING

This section describes how SQUID calibrates the recall model in §4.4. The calibration is an offline profiling procedure for the deployed ANN index configuration; it is not performed while planning or serving queries. Its goal is to estimate how the HNSW search depth required for Recall@ k changes with partition size and route selectivity.

Profiling samples. We construct representative vector collections at several cardinalities N , build an HNSW index for each collection using the same index family and construction parameters as the deployment, and select a fixed set of query vectors. For each collection, we generate controlled visibility predicates that retain

fractions ρ of the vectors, spanning clean routes ($\rho = 1$) and increasingly mixed routes. We then sweep a grid of ef_s values. Thus, each profiling point is identified by (N, ρ, k, ef_s) , rather than by a particular authorization policy or a particular partition plan.

For every query and profiling point, an exact filtered top- k search gives the authorized ground-truth result set G . The HNSW search under the same visibility predicate returns A , and the measured response is

$$\text{Rec}@k = \frac{|A \cap G|}{|G|}. \quad (12)$$

We average this quantity over the profiling queries. The profiler retains a measurement only when the database executes the approximate query through the HNSW index; samples for which the optimizer switches to a different execution strategy are outside the scope of this HNSW recall model.

Two-stage Hill fit. For every profiling point, we first normalize search effort by the expected number of visible candidates per requested result, $x = ef_s \rho / k$. For each sampled cardinality N , we fit the Hill curve

$$\text{Rec}(x; N) = \frac{x^{\gamma_N}}{\lambda_N^{\gamma_N} + x^{\gamma_N}} \quad (13)$$

to its HNSW measurements by bounded nonlinear least squares. The per-size fits estimate both the saturation exponent γ_N and the effort scale λ_N . SQUID uses the median of the fitted γ_N values as a single global exponent γ , refits one λ_N for each sampled size with γ fixed, and finally fits $\lambda_N = \lambda_0 + \lambda_1 \log(1 + N)$ by least squares. This produces the three parameters $\gamma, \lambda_0, \lambda_1$ used in the recall equation in §4.4.

Using the fitted model. Given a planning target τ , SQUID first inverts the fitted curve for a clean route to obtain its base search depth,

$$ef_s^{\text{base}}(N, k, \tau) = k\lambda(N) \left(\frac{\tau}{1 - \tau} \right)^{1/\gamma}. \quad (14)$$

For a route with selectivity $\rho(u, P)$, it uses $ef_s^{\text{base}}/\rho(u, P)$, clipped to the feasible range for the partition, as the recall-normalized search-depth estimate. Hence the model does not assume that a fixed ef_s gives comparable recall across layouts: a larger partition changes $\lambda(N)$, while a mixed route decreases the effective visible-candidate budget through $\rho(u, P)$. If the ANN index configuration, vector distribution, or target k changes materially, the same profiling procedure is rerun to recalibrate the model.

C PARTITION-PLANNING DETAILS

This appendix supplies the formal details omitted from Section 4. They justify the seed accounting, candidate sparsification, and coverage-preserving local updates used by the Planner; the main text retains the corresponding method definitions and control flow.

C.1 Decision Version and Complexity

The decision version of MBP² takes an Access Atom Catalog, a memory budget $\mathcal{B}$, a query-computation threshold $\mathcal{K}$, and the per-route costs $C(u, P)$. It asks whether there exists a partition plan $\mathcal{P}$

and routes $\{\mathcal{P}(u)\}_{u \in U}$ satisfying route coverage,

$$D_u \subseteq \bigcup_{P \in \mathcal{P}(u)} P \quad \text{for every } u \in U,$$

the copy budget $M(\mathcal{P}) \leq \mathcal{B}$, and $C(\mathcal{P}) \leq \mathcal{K}$. The following theorem establishes the hardness cited in Section 4.1.

THEOREM C.1. MBP² is NP-hard.

PROOF. Consider a restricted family of MBP² instances with n independent gadgets. Gadget i is parameterized by positive integers v_i and q_i and contains two users a_i, b_i . Their valid search domains consist of a shared Access Atom S_i of cardinality v_i and respective singleton private atoms X_i and Y_i . This planning instance admits the two-partition plan

$$P_{a_i} = X_i \cup S_i, \quad P_{b_i} = Y_i \cup S_i. \quad (15)$$

The construction specifies a valid instance of the general planning problem; it does not require SQUID's seed-generation procedure. The reduction uses the abstract per-route cost supplied with an MBP² instance. For $u \in \{a_i, b_i\}$, set

$$C(u, P) = \frac{\alpha_i}{\rho(u, P)}, \quad \alpha_i = \frac{q_i(v_i + 1)}{2}. \quad (16)$$

Thus, q_i is encoded in the route-cost table, rather than in a user or workload weight. Different item gadgets share neither users nor atoms.

Within gadget i , retaining two copies of S_i incurs no consolidation penalty, whereas materializing one copy saves exactly v_i vectors. The least expensive one-copy organization places X_i, S_i , and Y_i together. Each user then sees $v_i + 1$ of its $v_i + 2$ vectors, so consolidation increases query computation by

$$2\alpha_i \left(\frac{v_i + 2}{v_i + 1} - 1 \right) = q_i. \quad (17)$$

Absorbing S_i into either private side increases computation by $\alpha_i(v_i + 1)/v_i \geq q_i$, and isolating S_i increases it by $2\alpha_i \geq q_i$; the remaining reorganization is no cheaper. Thus, every other one-copy organization for this gadget has query-computation increase at least q_i .

Let $x_i \in \{0, 1\}$ indicate whether gadget i uses one shared copy, and let M_0 and C_0 denote the memory consumption and query computation of the two-copy organization. A plan satisfying memory budget $\mathcal{B}$ and decision threshold $\mathcal{K}$ must therefore satisfy

$$\sum_{i=1}^n v_i x_i \geq M_0 - \mathcal{B}, \quad \sum_{i=1}^n q_i x_i \leq \mathcal{K} - C_0. \quad (18)$$

It remains to show that the full planning space introduces no cheaper alternative. A partition mixing atoms from different gadgets cannot remove a copy beyond the copy removed within each gadget. Splitting every such partition along gadget boundaries preserves memory consumption and route coverage, and it weakly increases each route's selectivity; under $C(u, P) = \alpha_i/\rho(u, P)$, this transformation does not increase query computation. Thus, whenever a feasible plan exists, a gadget-separated feasible plan exists as well. The two inequalities are consequently the decision form of 0-1 Knapsack: saved memory is item value, additional query computation is item weight, $M_0 - \mathcal{B}$ is the target value, and $\mathcal{K} - C_0$ is the capacity. Setting $\mathcal{B} = M_0 - V$ and $\mathcal{K} = C_0 + W$ maps any

Knapsack target V and capacity W to the constructed planning instance in polynomial time and preserves feasibility. Therefore, MBP² is NP-hard. $\square$

C.2 Seed Copy Accounting

For each user u , the per-user seed materializes one partition $P_u^{(0)} = D_u$. Hence, every Access Atom S_{A_i} is copied once for every user in its visible-user set A_i , and the seed consumes

$$M(\mathcal{P}^{(0)}) = \sum_{A_i \in \mathcal{A}} |A_i| |S_{A_i}|.$$

Equivalently, summing the vectors in each user-visible domain gives

$$M(\mathcal{P}^{(0)}) = \sum_{u \in U} |D_u|.$$

The equality follows because each vector belongs to exactly one Access Atom and contributes to precisely the domains of the users in that Atom's visible-user set.

C.3 Sparse Shared-Atom Candidates

For an Atom S_{A_i} with $r = |\mathcal{I}(S_{A_i})|$ owner partitions, the complete owner clique contains $\binom{r}{2}$ edges. The core-star keeps only the $r - 1$ edges between $\text{core}(S_{A_i})$ and every other owner. Thus every non-core copy appears on one retained edge, while the atom-induced graph size decreases from $O(r^2)$ to $O(r)$. The omitted edges represent alternative pairwise consolidation orders, so this construction deliberately sparsifies rather than exhausts the candidate space.

When several Atoms induce the same partition pair, their shared-vector masses are accumulated in $\Gamma(P_x, P_y)$. SQUID retains all core-star edges and, for each partition, retains the top- d remaining incident edges according to $\phi(P_x, P_y)$. Ties are resolved deterministically by partition identifiers; an adjacency list or priority queue can maintain these candidates without changing the planning rule.

C.4 Coverage-Preserving Route Updates

Consider a valid local transformation of edge (P_x, P_y) with output partitions $\mathcal{T}_o(P_x, P_y)$. For every user whose route contains an endpoint, the Planner removes the endpoint partitions and adds each output partition that contains an Access Atom contributed by those endpoints to that user's old coverage. Routes that contain neither endpoint are unchanged.

Every transformation in Section 4.3 preserves the Atom union,

$$\bigcup_{Q \in \mathcal{T}_o(P_x, P_y)} \mathcal{S}(Q) = \mathcal{S}(P_x) \cup \mathcal{S}(P_y).$$

Thus, every Atom reachable through either old endpoint remains reachable through at least one selected output partition, proving route coverage after the replacement. The transformation changes placements only; it does not change any Atom's visible-user set. Therefore, coverage ensures physical reachability and the Search Engine's visibility filtering preserves authorization correctness, including when an output partition contains Atoms visible to different users.

C.5 Local Transformation Accounting

For a retained edge (P_x, P_y) , let $\Omega = \mathcal{S}(P_x) \cap \mathcal{S}(P_y)$, $X = \mathcal{S}(P_x) \setminus \Omega$, and $Y = \mathcal{S}(P_y) \setminus \Omega$. These three Atom sets are pairwise disjoint. Let $\mathcal{T}_o(P_x, P_y)$ denote the nonempty output partitions of any transformation o in Section 4.3. Every such transformation satisfies

$$\bigcup_{Q \in \mathcal{T}_o(P_x, P_y)} \mathcal{S}(Q) = X \cup \Omega \cup Y \quad (19)$$

and materializes every Atom in Ω , X , and Y exactly once. Relative to the two input endpoints, the immediate memory release is therefore

$$\Delta M_o = \sum_{S_{A_i} \in \Omega} |S_{A_i}| > 0. \quad (20)$$

Only routes containing an endpoint can change. Let

$$U_e = \{u \in U \mid P_x \in \mathcal{P}(u) \vee P_y \in \mathcal{P}(u)\}. \quad (21)$$

After applying the coverage-preserving replacement rule, the corresponding query-computation change can be evaluated locally as

$$\Delta C_o = \sum_{u \in U_e} \left(\sum_{Q \in \mathcal{P}_o(u) \cap \mathcal{T}_o(P_x, P_y)} C(u, Q) - \sum_{Q \in \mathcal{P}(u) \cap \{P_x, P_y\}} C(u, Q) \right). \quad (22)$$

All route contributions outside U_e cancel because their partitions and routes are unchanged. This locality permits the Planner to refresh only transformations incident to the newly materialized outputs.

D INCREMENTAL-MAINTENANCE DETAILS

This section formalizes the invariants and transition rules maintained by the update procedure in Section 5.2.

D.1 Published-State Invariants

Let a published serving state be

$$\Sigma = (\mathcal{V}, \mathcal{A}, \mathcal{P}, \{\mathcal{P}(u)\}_{u \in U}, \mathcal{I}, \mathcal{G}), \quad (23)$$

where $\mathcal{A}$ is the Access Atom Catalog, $\mathcal{I}$ the Atom Inverted Table, and $\mathcal{G}$ the Shared-Atom Graph; per-partition indexes are associated with $\mathcal{P}$. A batch changes Σ to Σ' . SQUID publishes Σ' only after the following conditions hold:

$$d \in S'_{A_i} \iff \mathcal{V}'(d) = A_i, \quad \forall d \in D', \quad (24)$$

$$D'_u \subseteq \bigcup_{P \in \mathcal{P}'(u)} P, \quad \forall u \in U, \quad (25)$$

$$M(\mathcal{P}') \leq \mathcal{B}. \quad (26)$$

Equation (24) gives Atom consistency, while Equations (25) and (26) give physical reachability and budget feasibility. Authorization safety is enforced against $\mathcal{V}'$ during candidate filtering, so every returned vector belongs to D'_u . New partitions and indexes are materialized before their routes become visible; obsolete copies are reclaimed only after the replacement state is published.

D.2 Local Repair Scope

Let $\mathcal{P}_\Delta$ contain partitions directly modified, created, or retired by a batch. The efficiency-repair scope is

$$\mathcal{P}_\Delta^+ = \mathcal{P}_\Delta \cup \mathcal{N}_G(\mathcal{P}_\Delta). \quad (27)$$

The one-hop extension includes partitions whose consolidation opportunities can change because they share an Atom with the directly affected state. Candidate transformations are regenerated only for edges incident to this region. Partitions outside the region retain their Atom placements and indexes, and routes not covering a changed Atom retain their entries.

D.3 Compaction and Split Semantics

For a partition P , let $L(P)$ and $T(P)$ be its live and inactive vector copies. Compaction is triggered when

$$\delta(P) = \frac{|T(P)|}{|L(P)| + |T(P)|} > \eta. \quad (28)$$

It replaces the physical contents and index of P with those induced by $L(P)$. Because it changes neither the live-Atom placements nor the route entries, compaction preserves Equations (24) and (25) while reducing physical storage.

A selectivity split replaces a live partition P by outputs P_1, P_2 satisfying

$$\mathcal{S}(P_1) \cup \mathcal{S}(P_2) = \mathcal{S}(P), \quad (29)$$

where $\mathcal{S}(P_1) \cap \mathcal{S}(P_2)$ contains only Atoms intentionally replicated for both output route groups. The route-replacement rule in Appendix C.4 preserves coverage. SQUID accepts the split only if its resulting organization $\mathcal{P}'$ satisfies $C(\mathcal{P}') < C(\mathcal{P})$ and $M(\mathcal{P}') \leq \mathcal{B}$.

E PARTITION-ORGANIZATION CASE STUDY

We examine the partition organizations produced on SIFT+Tree-based at the approximately 1.5 $\times$ vector-copy setting. For each routed user-partition pair, $\rho(u, P)$ is the fraction of vectors in P visible to u . We report its mean, s , across routed pairs and the mean number p of partitions on a user route. Higher s indicates cleaner routed search spaces, while lower p avoids distributing a query across additional indexes.

RLS and ROLE are reference endpoints rather than budget-comparable methods: RLS searches one shared partition, while ROLE fully materializes isolated role-local partitions. Figure 12 places the remaining methods between these endpoints; its dashed curve connects the non-dominated budget-comparable points and excludes the unconstrained ROLE reference. The footprint column shows the storage cost omitted from the two-dimensional plot. RLS keeps only one shared index (1.418 GB) but has very low selectivity ($s = 0.036$); ROLE attains pure, single-partition routes only by increasing the footprint to 4.994 GB.

SQUID reaches $s = 0.830$ with $p = 1.62$ and a 2.115 GB footprint. Relative to HONEYBEE, it raises selectivity from 0.563 to 0.830 for essentially the same footprint (2.117 versus 2.115 GB), at the cost of 0.56 more routed partitions per user. Relative to VEDA, SQUID improves all three reported properties: it has higher selectivity (0.830 versus 0.687), lower fan-out (1.62 versus 1.89), and lower footprint (2.115 versus 2.241 GB). EFFVEDA and HQI reach pure routes only by routing users to substantially more partitions, while ROLE reaches the isolation endpoint through unconstrained replication. The case study therefore complements the end-to-end measurements in Section 6.3: SQUID’s performance arises from a partition organization that retains selective routes without broadly increasing per-query route fan-out or realized footprint.

F TIME-COMPLEXITY ANALYSIS

This appendix analyzes the logical work performed by SQUID. We distinguish it from physical ANN-index construction: the Planner determines Atom placements and routes, whereas materializing a final partition also invokes the chosen index implementation. Let $T_{\text{build}}(N)$ and $T_{\text{ann}}(N, ef)$ denote, respectively, the construction and query costs of the selected ANN index on N vectors at search depth ef . This makes the analysis applicable to HNSW as used in our implementation as well as to compatible per-partition indexes.

F.1 Notation

Let $n = |D|$, $m = |\mathcal{A}|$, and $p = |\mathcal{P}|$ denote the numbers of vectors, Access Atoms, and current partitions, respectively. The resolved-visibility relation contains

$$L = \sum_{d \in D} |\mathcal{V}(d)| = \sum_{S_{A_i} \in \mathcal{A}} |A_i| |S_{A_i}|$$

vector-user incidences. For Atom S_{A_i} , let $r_i = |\mathcal{I}(S_{A_i})|$ be its number of owner partitions, and let

$$R = \sum_{S_{A_i} \in \mathcal{A}} r_i$$

be the number of Atom placements. Thus R measures logical replication, while $M(\mathcal{P}) = \sum_i r_i |S_{A_i}|$ measures the corresponding vector-copy budget.

Let $H = \sum_{u \in U} |\mathcal{P}(u)|$ be the number of route-partition memberships. For a retained Shared-Atom Graph edge $e = (P_x, P_y)$, define

$$\kappa_e = |\mathcal{S}(P_x) \cup \mathcal{S}(P_y)| + |U_e|, \quad U_e = \{u \mid P_x \in \mathcal{P}(u) \vee P_y \in \mathcal{P}(u)\}.$$

The first term is the Atom state needed to form X, Ω, Y ; the second is the number of routes whose local cost or coverage may change. Cardinalities, owner lists, route memberships, and per-route visible counts are maintained in the catalog, inverted table, and routing directory. Consequently, a recall-model evaluation for one routed output partition is $O(1)$, and evaluating all five alternatives for e takes $O(\kappa_e)$ time. The constant five does not appear in the asymptotic bounds.

F.2 Atomization and Seed Construction

If every vector’s resolved visible-user set is supplied in canonical form, hashing that set and inserting the vector into its Access Atom takes expected $O(L+n)$ time and $O(L+n+m)$ space. A deterministic implementation that sorts the n visibility signatures instead has $O(L + n \log n)$ time. These bounds include reading the resolved authorization result; SQUID does not inspect the policy program that produced it.

The per-user seed creates one logical placement of S_{A_i} for every user in A_i . Its placement count is

$$R_0 = \sum_{S_{A_i} \in \mathcal{A}} |A_i|,$$

so constructing the seed’s route and inverted-table entries costs $O(R_0 + |U|)$. Physically materializing the seed writes $L = M(\mathcal{P}^{(0)})$ vector copies and therefore costs $O(L)$, excluding index construction. If the seed indexes are built immediately, the additional cost is $\sum_{P \in \mathcal{P}^{(0)}} T_{\text{build}}(|P|)$.

F.3 Shared-Atom Graph Construction

For each Atom with $r_i > 1$, core-star construction scans its owner list once to select the core and emits $r_i - 1$ raw star incidences. Hence the number of raw incidences is

$$E_{\text{raw}} = \sum_{i=1}^m \max\{r_i - 1, 0\} = O(R),$$

and building the owner lists, core choices, and aggregated shared-Atom sets costs $O(R + E_{\text{raw}})$ time. The graph is subsequently pruned to the top- d incident star edges per partition. Sorting the raw incident lists costs

$$O\left(\sum_{P \in \mathcal{P}} \deg_{\text{raw}}(P) \log \deg_{\text{raw}}(P)\right) \subseteq O(E_{\text{raw}} \log p),$$

and the retained graph has

$$|E| \leq \min\{E_{\text{raw}}, dp\}.$$

Thus the candidate graph uses $O(R + |E|)$ space rather than the $O(\sum_i r_i^2)$ space of materializing every Atom-owner clique.

For every retained edge, SQUID enumerates the five local transformations, retains the best memory-saving alternative under the cost-priority rule, and pushes it into a heap. Initial candidate evaluation and heap construction therefore cost

$$T_{\text{init-heap}} = O\left(\sum_{e \in E} \kappa_e + |E| \log |E|\right).$$

This bound is output-sensitive: an Atom shared by many partitions contributes linearly through its core star, while the top- d rule bounds the number of costly local evaluations that reach the heap.

F.4 Greedy Partition Planning

Let q be the number of accepted consolidations before the copy budget is met. Every accepted offline transformation removes at least one Atom placement and does not introduce a new placement, so $q \leq R_0 - m$. At iteration t , let F_t be the number of retained edges revalidated after replacing the two endpoint partitions, and let $\bar{\kappa}_t$ be the average local-evaluation size among those edges. Popping from the heap, applying the route-preserving replacement, and updating the affected catalog and route entries cost $O(\kappa_t + \log |E|)$. Refreshing the changed neighborhood costs $O(F_t(\bar{\kappa}_t + \log |E|))$. The logical planning time is therefore

$$T_{\text{plan}} = O\left(L + R + E_{\text{raw}} \log p + \sum_{e \in E} \kappa_e + |E| \log |E|\right) + O\left(\sum_{t=1}^q [\kappa_t + \log |E| + F_t(\bar{\kappa}_t + \log |E|)]\right).$$

Only edges incident to changed output partitions are refreshed; unrelated partitions, routes, and heap entries are retained. A defensive heap rebuild, used only after stale entries exhaust the heap, reconstructs the current core-star graph and has the same asymptotic cost as graph construction plus $T_{\text{init-heap}}$. If b_{rb} such rebuilds occur, the displayed bound gains $b_{\text{rb}}(O(R + E_{\text{raw}} \log p) + T_{\text{init-heap}})$. In the normal feasible-budget path, local refreshes maintain the candidate heap and this fallback is not on the critical path.

In the pessimistic case, a local transformation can affect all retained edges and all affected routes. Taking $F_t = O(|E|)$ and $\kappa_t, \bar{\kappa}_t = O(m + |U|)$ gives a conservative worst-case bound of

$$O(q|E|(m + |U| + \log |E|))$$

beyond initialization. This is intentionally loose: it describes a globally coupled instance, whereas SQUID's sparse graph and incremental refresh are designed to make $F_t \ll |E|$ in practice.

After the logical plan is fixed, physically building its ANN indexes costs

$$T_{\text{materialize}} = O(M(\mathcal{P})) + \sum_{P \in \mathcal{P}} T_{\text{build}}(|P|).$$

The first term writes vector copies, and the second is index-specific. It is reported separately from planning because it is not an operation performed by the Shared-Atom Graph or the recall-aware cost model.

F.5 Query-Time Cost

For a query from user u , let $h_u = |\mathcal{P}(u)|$. The Search Engine first sorts its route by stored selectivity, costing $O(h_u \log h_u)$. It then probes each routed partition once at ef_0 , and may issue at most one expanded search at ef_p . With constant-time visibility checks and a size- k result heap, the query cost is bounded by

$$T_{\text{query}}(u) = O\left(h_u \log h_u + \sum_{P \in \mathcal{P}(u)} [T_{\text{ann}}(|P|, ef_0) + ef_0 \log k + \mathbf{1}_P(T_{\text{ann}}(|P|, ef_p) + ef_p \log k)]\right),$$

where $\mathbf{1}_P \in \{0, 1\}$ indicates whether the route is expanded. The $ef \log k$ terms account for authorization filtering and merging candidates into the global top- k . The expression deliberately leaves T_{ann} abstract because exact ANN search complexity depends on the selected index and its parameters; SQUID adds only route ordering, filtering, and result merging.

F.6 Incremental-Maintenance Cost

Consider one update batch. Let L_{Δ} be the number of changed vector-user visibility incidences, R_{Δ} the Atom placements touched by the affected Atoms, and U_{Δ} the number of affected routes. Updating resolved visibility, the Atom Catalog, the Atom Inverted Table, and the User Routing Directory costs

$$O(L_{\Delta} + R_{\Delta} + U_{\Delta}),$$

excluding index insertion or deletion work. The one-hop repair region contains $p_{\Delta}^+ = |\mathcal{P}_{\Delta}^+|$ partitions and E_{Δ} retained Shared-Atom edges. If q_{Δ} local consolidations are accepted and $F_{\Delta,t}$ edges are refreshed after step t , its logical repair cost is

$$T_{\text{repair}} = O\left(\sum_{e \in E_{\Delta}} \kappa_e + |E_{\Delta}| \log |E_{\Delta}| + \sum_{t=1}^{q_{\Delta}} [\kappa_t + \log |E_{\Delta}| + F_{\Delta,t}(\bar{\kappa}_t + \log |E_{\Delta}|)]\right).$$

This has the same form as offline planning, but depends only on the changed one-hop neighborhood rather than the global graph.

Compaction checks tombstone ratios in $O(p_\Delta^+)$ time. If C_Δ is the set of compacted partitions, rebuilding their live-vector indexes adds

$$\sum_{P \in C_\Delta} (O(|P|) + T_{\text{build}}(|P|)).$$

For selectivity repair, let $a_P = |\mathcal{S}(P)|$, u_P be the number of users routed to a mixed partition P , and $\sigma_P = \sum_{S_{A_i} \in \mathcal{S}(P)} |A_i|$ be its Atom-user incidence count. Computing filtering impacts uses $O(\sigma_P)$ time. SQUID ranks users and considers $g = \min\{u_P, 64\}$ high-impact seed users in the implementation; testing seed pairs, assigning remaining users, and evaluating the resulting split take

$$O(\sigma_P + u_P \log u_P + g^2 a_P + u_P a_P).$$

The fixed cap makes the seed-pair term linear in a_P in the implementation. An accepted split additionally writes replicated copies and builds the two output indexes, costing $O(|P_1| + |P_2| + T_{\text{build}}(|P_1|) + T_{\text{build}}(|P_2|))$.

Combining these terms, the non-index portion of a batch is

$$O(L_\Delta + R_\Delta + U_\Delta + T_{\text{repair}} + p_\Delta^+ + \sum_{P \in \mathcal{S}_\Delta} T_{\text{split}}(P)),$$

where $\mathcal{S}_\Delta$ is the set of partitions considered for splitting. Physical index work is added only for new placements, compacted partitions, and accepted split outputs. Thus, unchanged partitions avoid both global candidate regeneration and index rebuilding.

F.7 Space Complexity

The persistent logical state consists of the resolved relation and Atom memberships, the Atom Inverted Table, user routes, and the retained graph. Its space is $O(L + n + m + R + H + |E|)$, in addition to the $O(M(\mathcal{P}))$ physical vector-copy and ANN-index footprint. The planner's candidate heap and edge cache require $O(|E|)$ live entries, with additional transient stale entries reclaimed by heap rebuilding. For an update batch, the auxiliary state is $O(R_\Delta + |E_\Delta| + U_\Delta)$ plus the materialized replacement partitions, rather than a duplicate of the full partition organization.